



The M-CATs

Biology Chapter 4: The Nervous System

4.0

Introduction to Nervous System

Roles of Nervous System

Movement



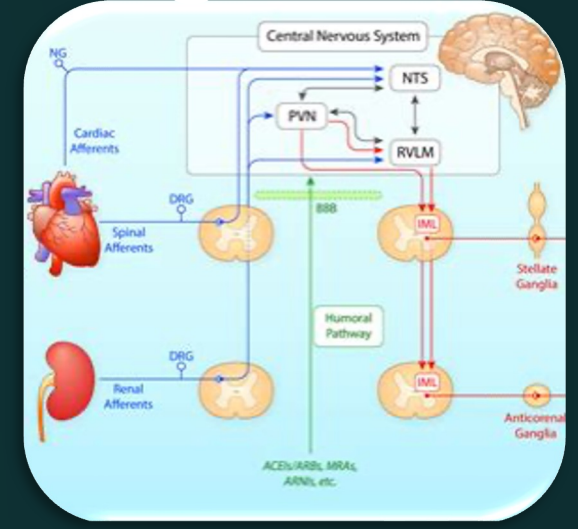
Sensation



Thought



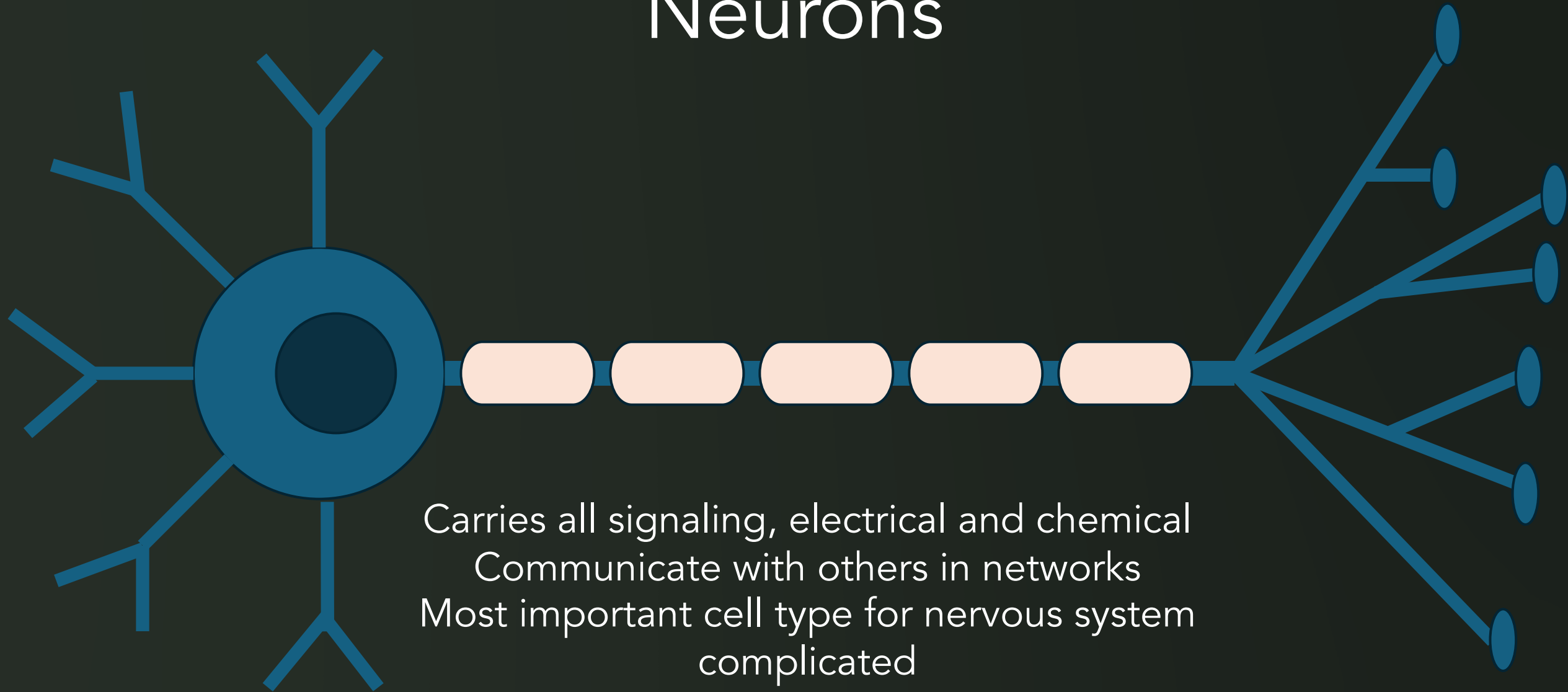
Organ Regulation



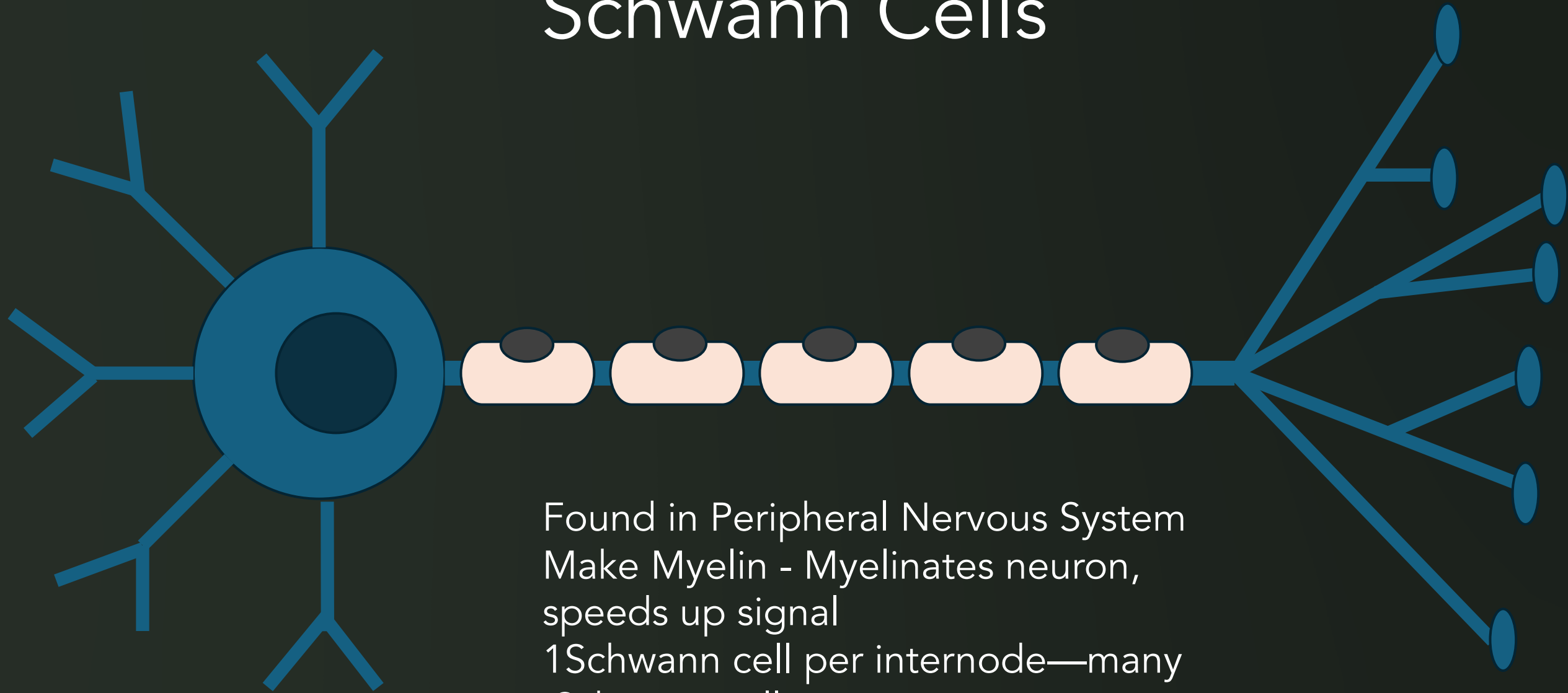
4.1

Cells of the Nervous System

Neurons



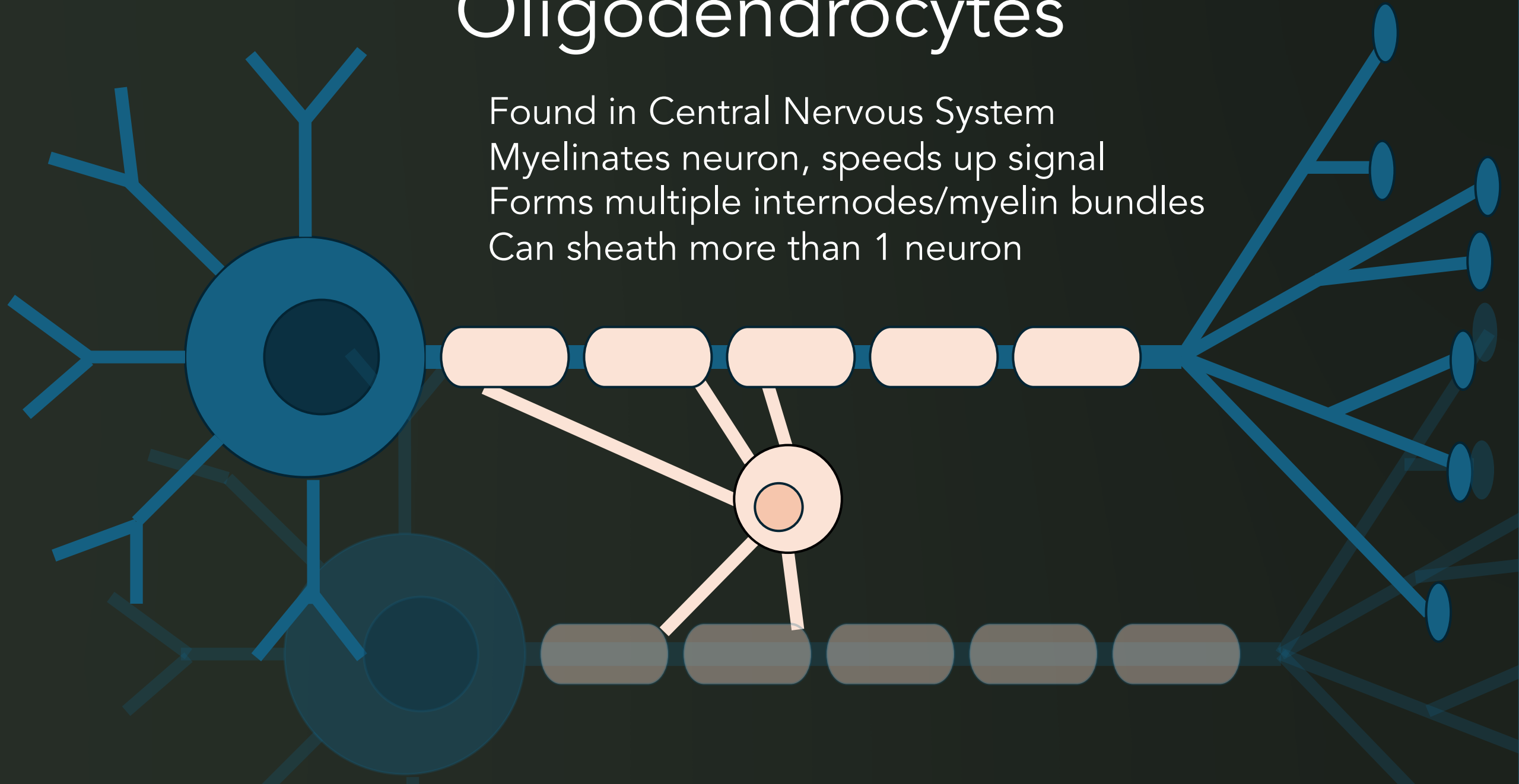
Schwann Cells



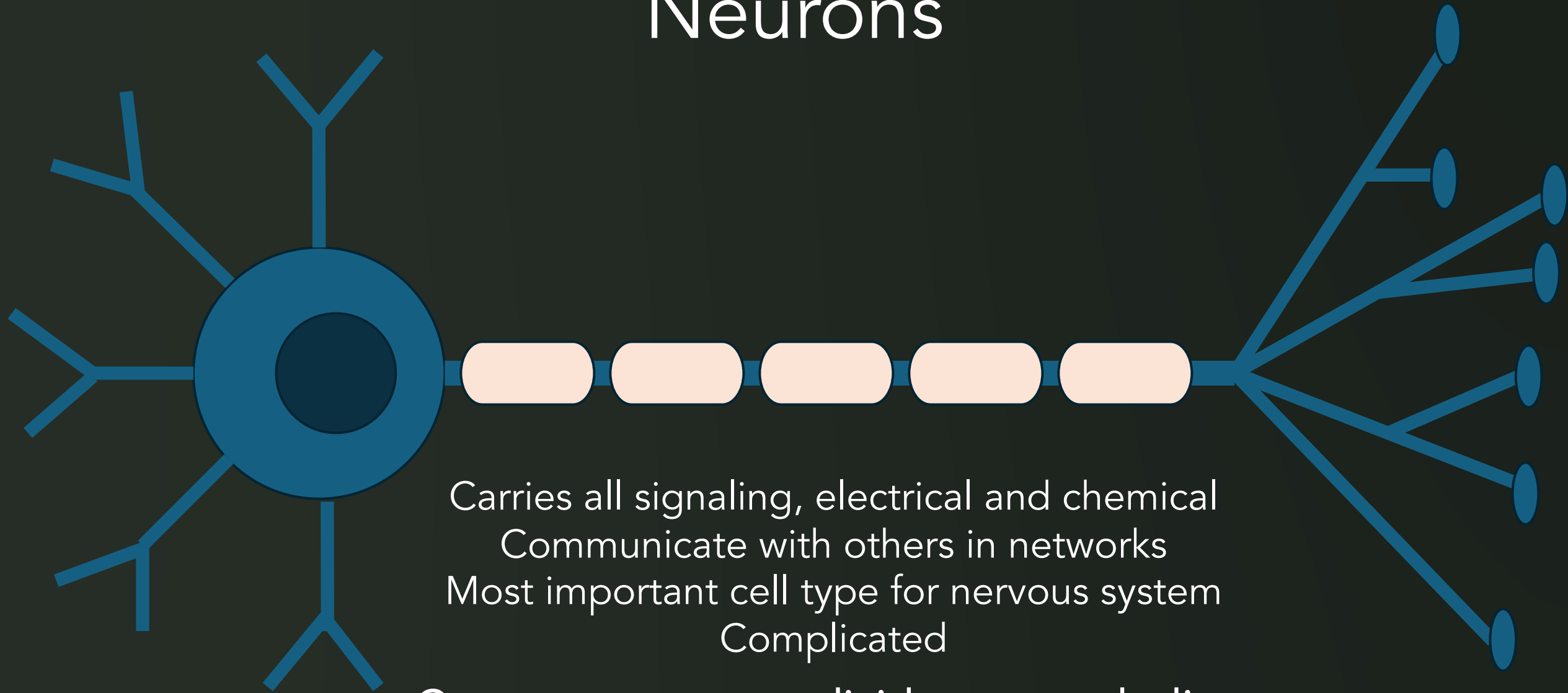
Found in Peripheral Nervous System
Make Myelin - Myelinates neuron,
speeds up signal
1 Schwann cell per internode—many
Schwann cells per neuron

Oligodendrocytes

Found in Central Nervous System
Myelinates neuron, speeds up signal
Forms multiple internodes/myelin bundles
Can sheath more than 1 neuron



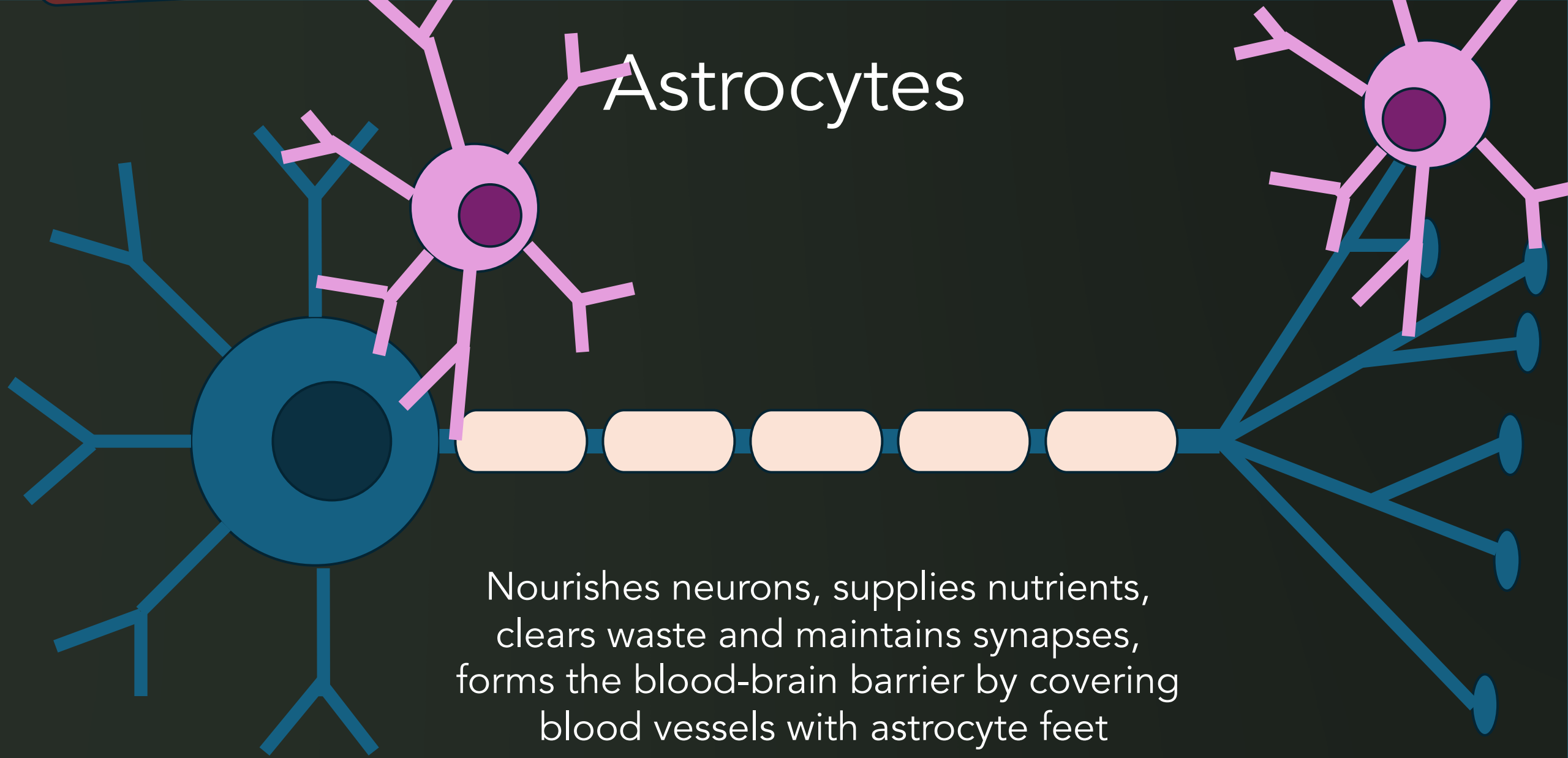
Neurons



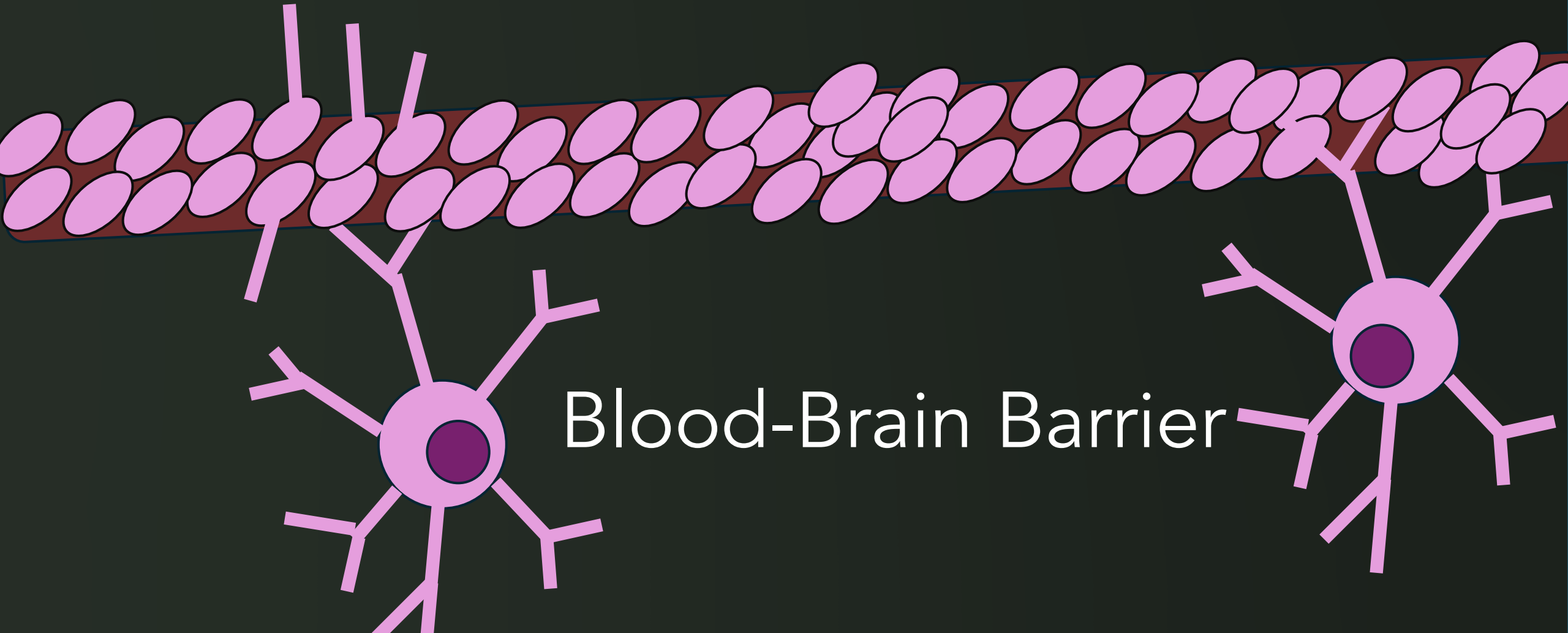
Carries all signaling, electrical and chemical
Communicate with others in networks
Most important cell type for nervous system
Complicated

Cannot regenerate, divide, or metabolize
completely

Astrocytes



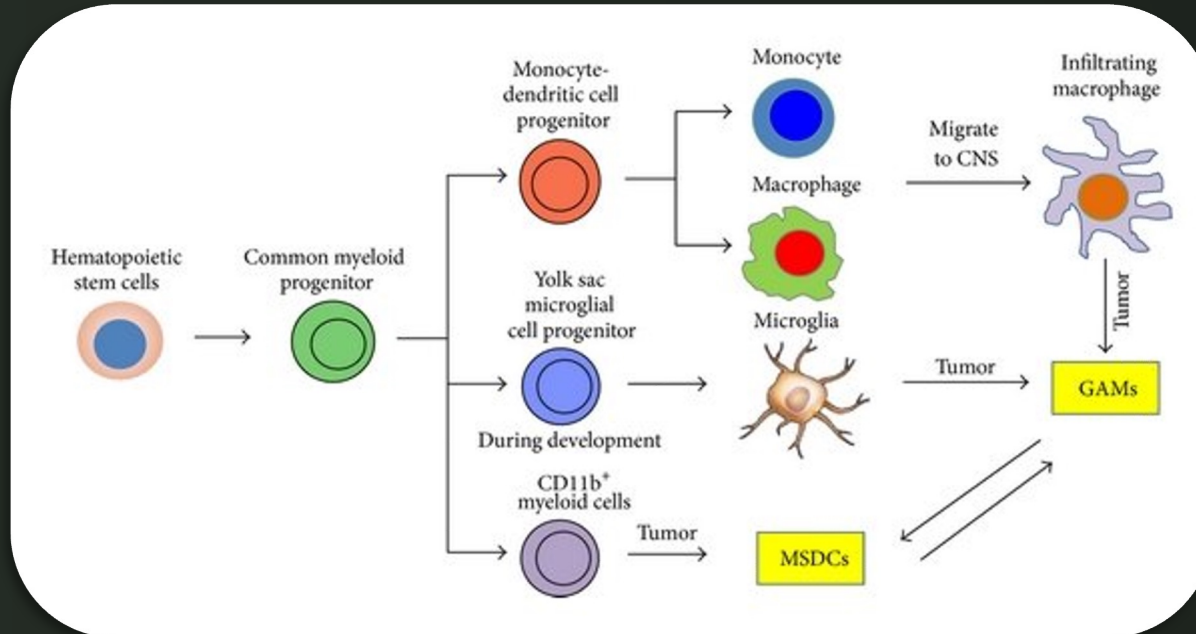
Highly regulated nutrient/solute exchange,
helpful for keeping pathogens out of the brain
specifies what passes through (lipophilic molecules,
carrier-specific molecules intended to pass through)



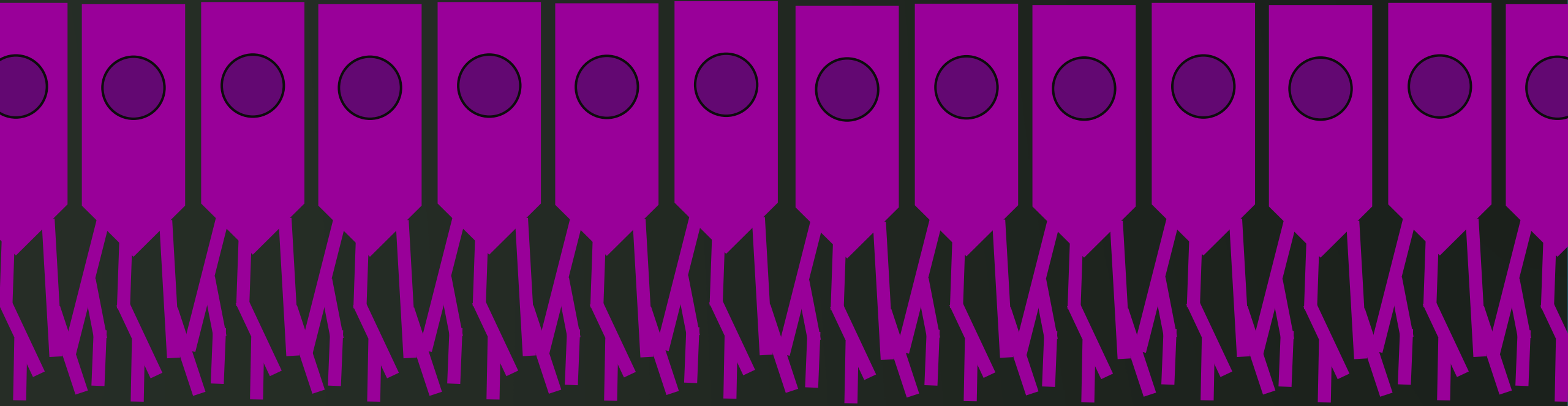
Microglia

Ingest and break down waste products, pathogens, and foreign objects through phagocytosis

Same family as macrophages and monocytes!



Ependymal Cells



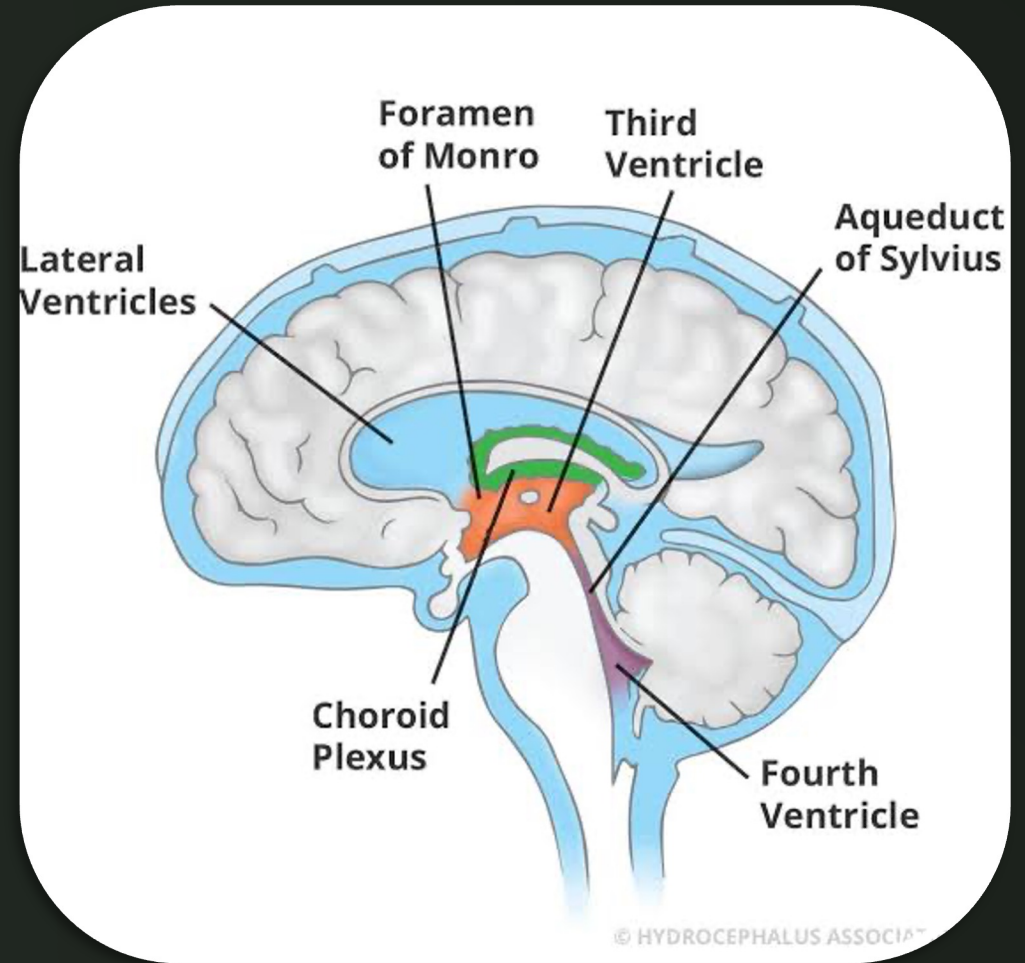
Line the ventricles of the brain,
produce cerebrospinal fluid (CSF)
Cilia on apical (inner-ventricle facing)
surface help circulate CSF

(Ventricular Space)

Cerebrospinal Fluid

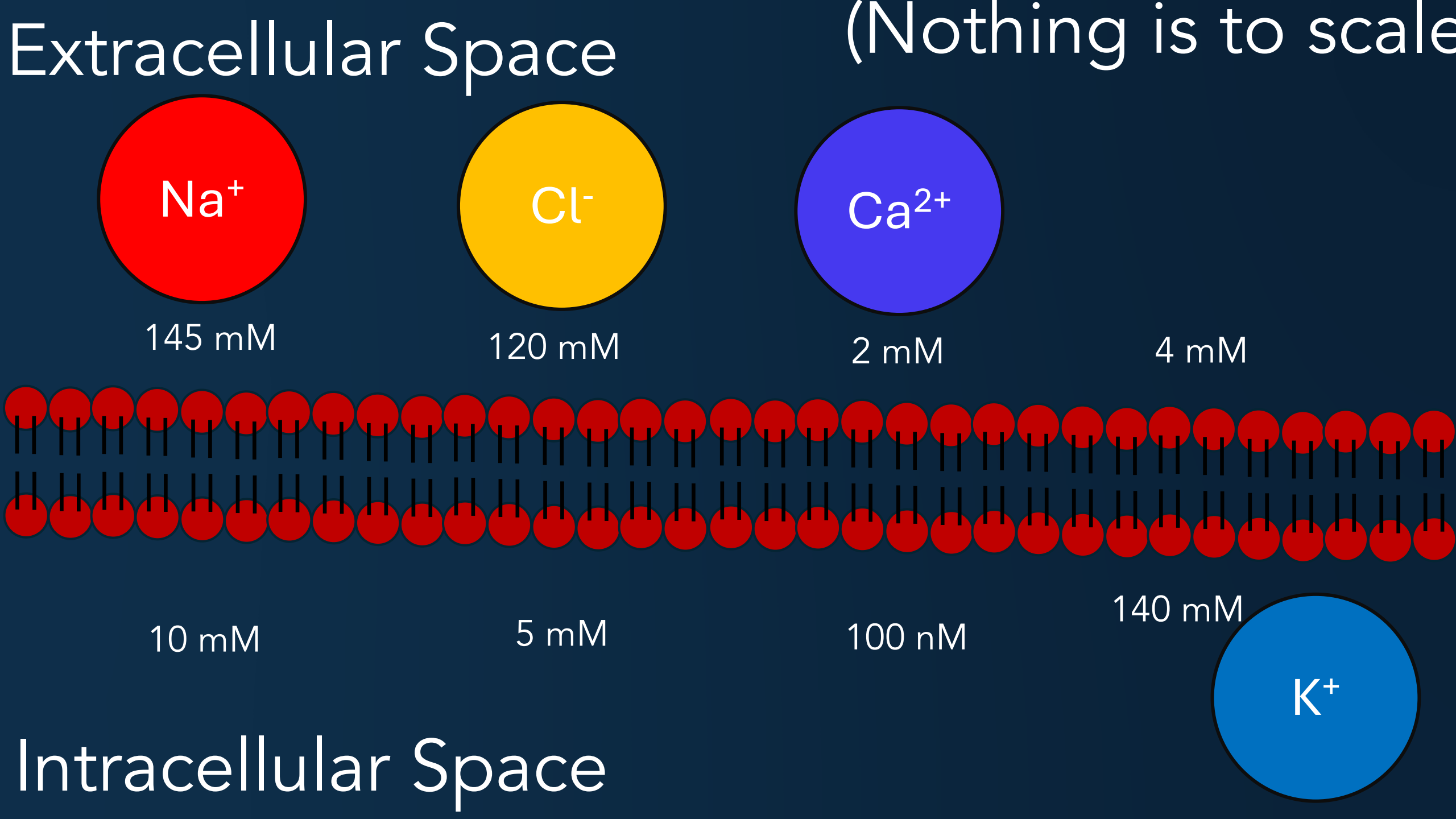
The brain floats in CSF and is cushioned by it. The brain weighs 1.4 kg, but buoyancy on CSF gives the brain a net weight of...

25 to 50 grams



4.2

Neural Potentials

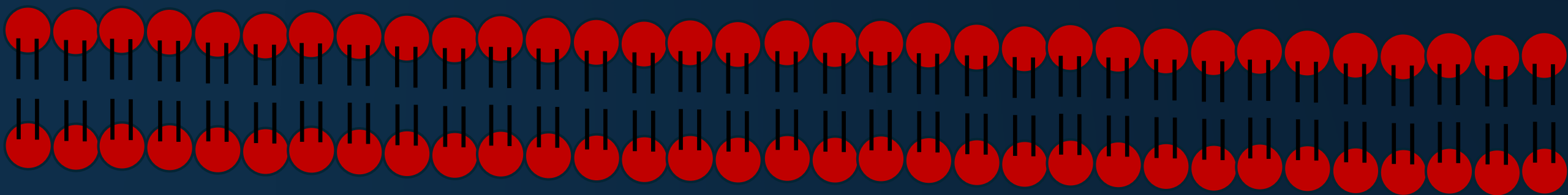


Membrane Potential

A measure of the stored charge inside the cell
relative to the outside of the cell

Positive membrane potential implies positive
charges are mostly outside of the cell

Neurons normally hold a negative membrane potential (-70 mV)



Membrane Potential

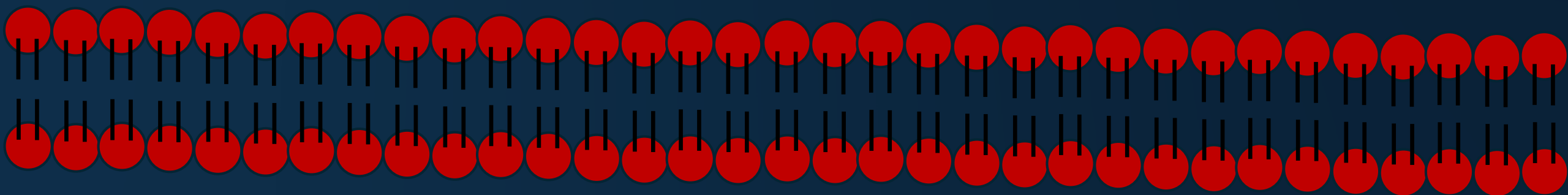
Depolarization is a change in membrane potential towards 0 mV

Since neurons rest at -70 mV, depolarization means an INCREASE in membrane potential

Hyperpolarization is a change in membrane potential away from 0 mV

Hyperpolarization refers to a DECREASE in membrane potential

Repolarization refers to restoration of membrane potentials. Neurons will usually depolarize then hyperpolarize. So, repolarization means the same as hyperpolarization.



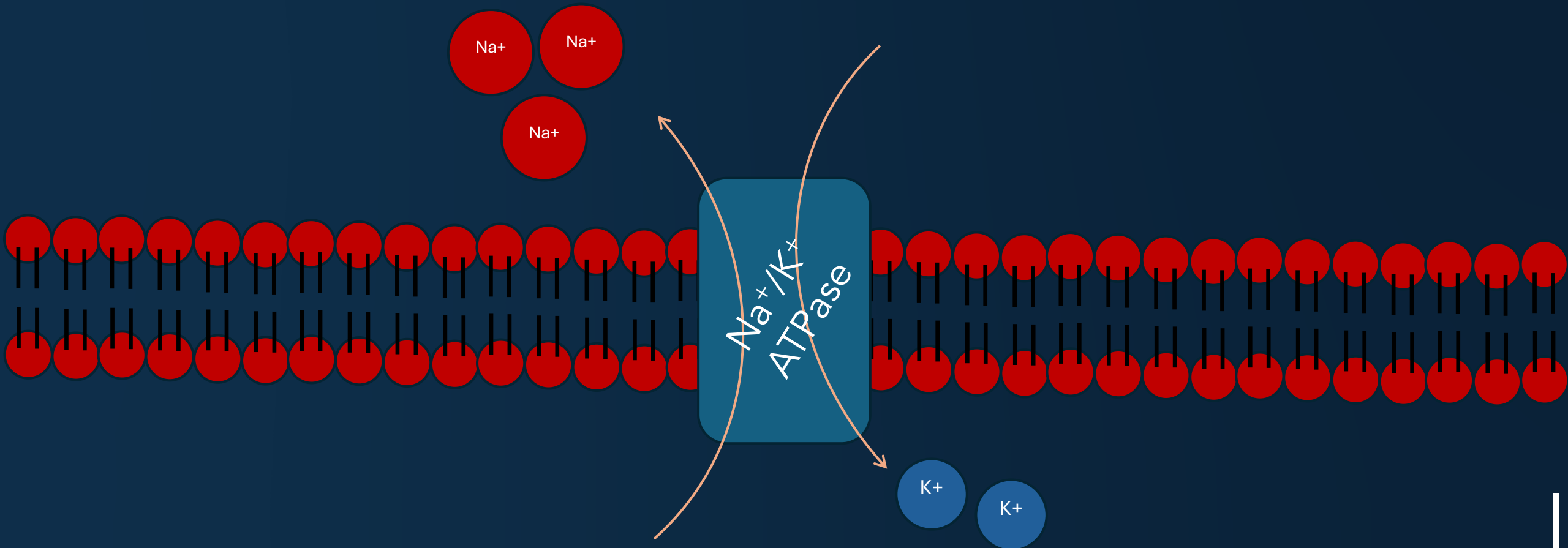
STRONG DEPOLARIZATION WILL ACTIVATE NEURON

Na⁺/K⁺ ATPase

Moves 3 Na⁺ out of the cell and 2 K⁺ into the cell with each ATP used up.

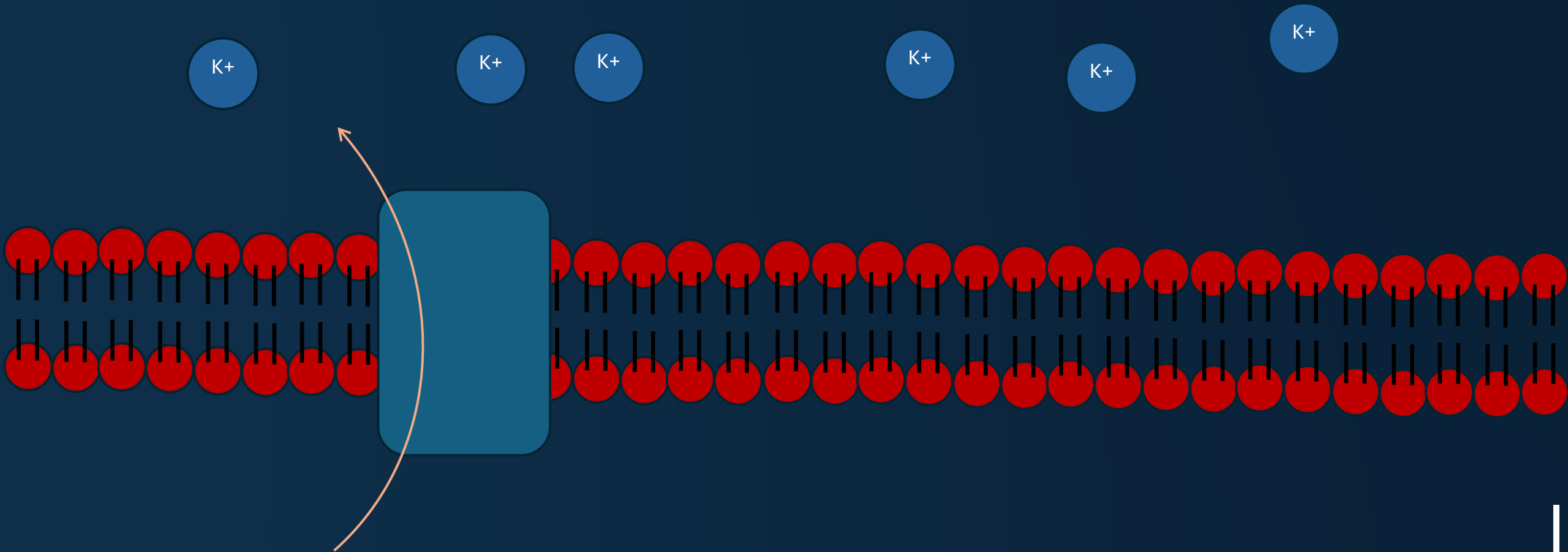
This pump is essential for generating and restoring the membrane potential.

This pump is found everywhere on the neuron membrane, but is densest on the axons and axon terminal.



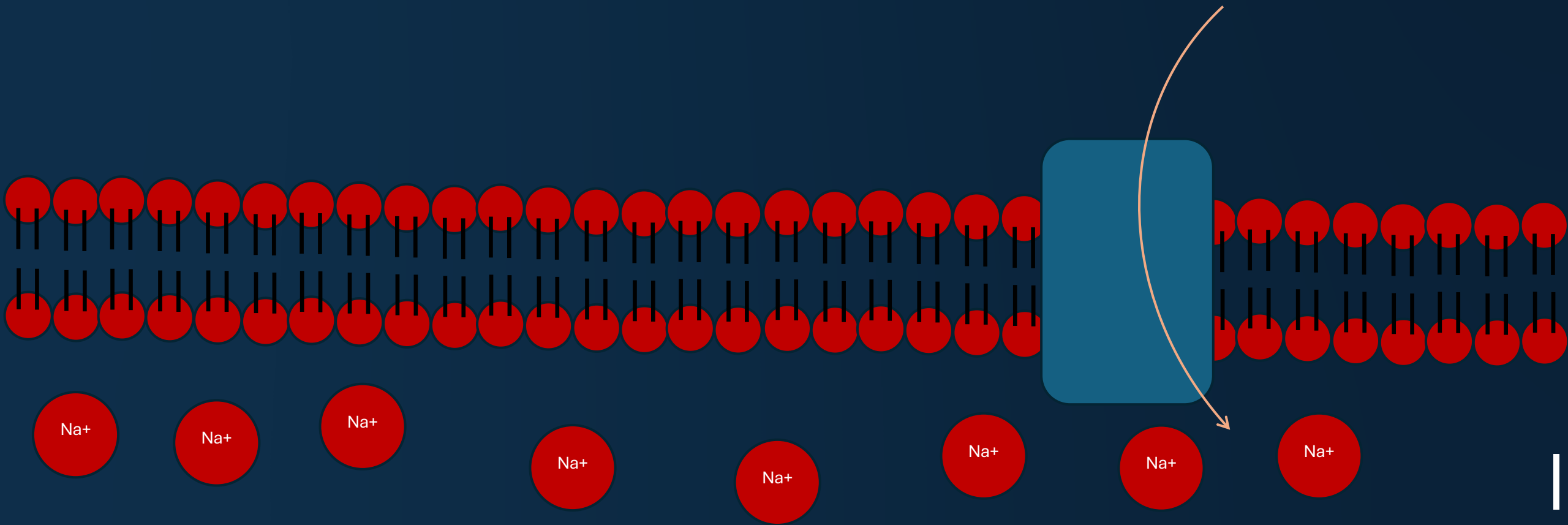
K⁺ Leak Channels

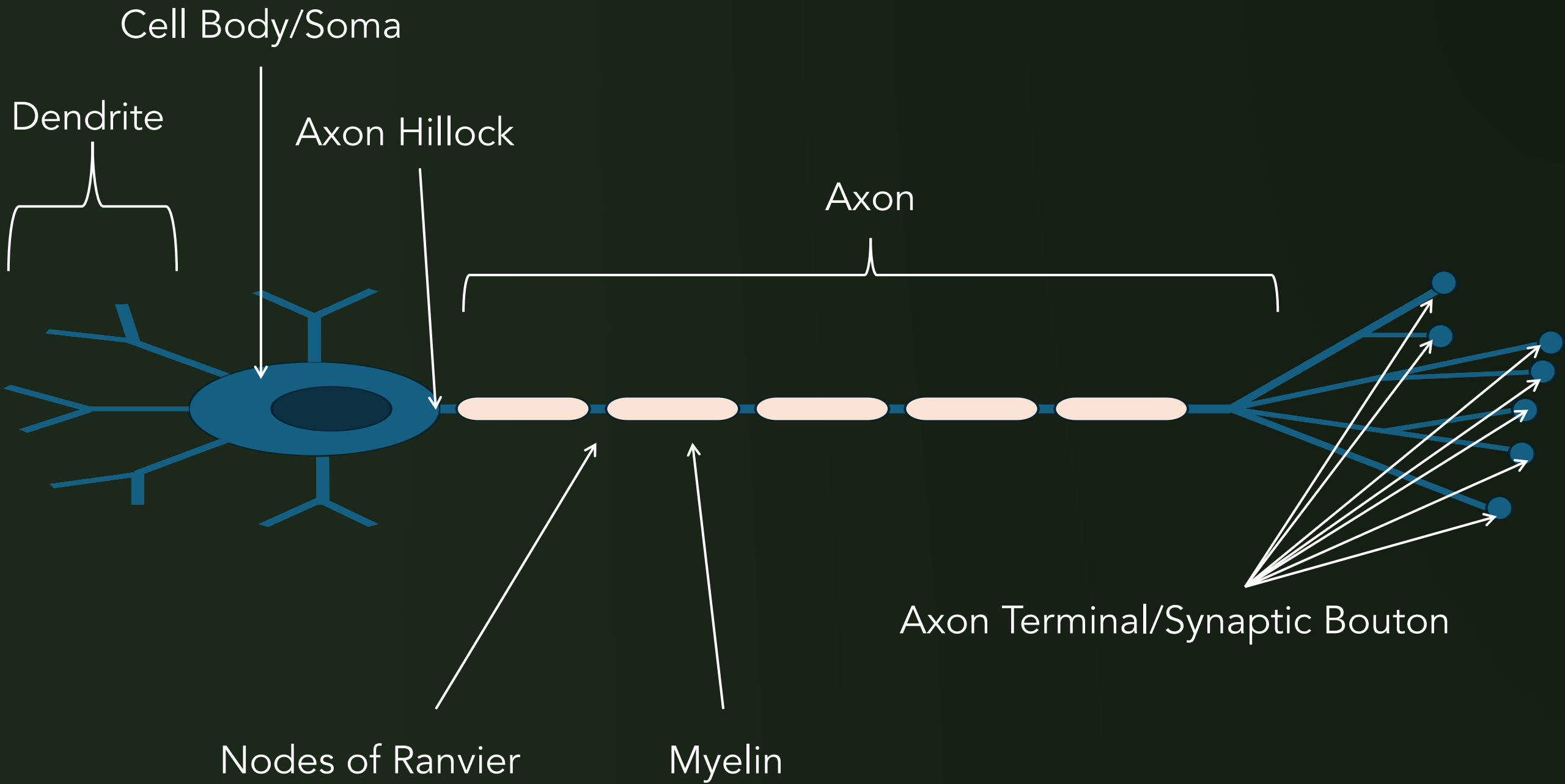
K⁺ can pass back; the membrane is MOST permeable to K⁺
Resting membrane potential lies near the potential of most permeable ion
These leak channels establish the -70 mV potential



Na⁺ Leak Channels

Not common, not a major contributor to membrane potential





Graded Potentials

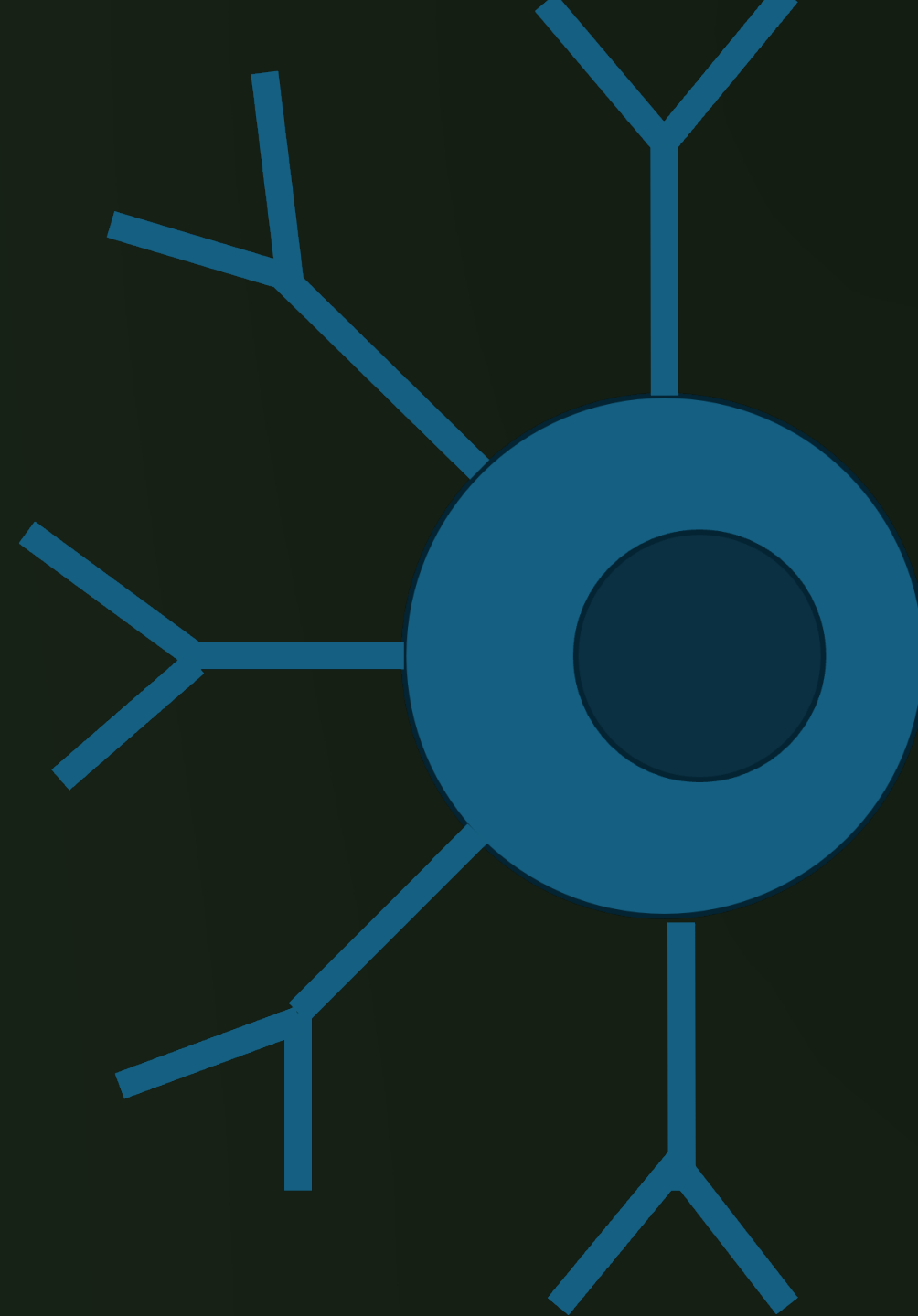
@dendrites + soma

There are ligand-gated channels on the cell membrane. The presence of a specific ligand opens channel

Flow of ions may depolarize or hyperpolarize cell

Depolarizations are called "excitatory post-synaptic potentials" or EPSPs

Depolarizations are called "inhibitory post-synaptic potentials" or IPSPs



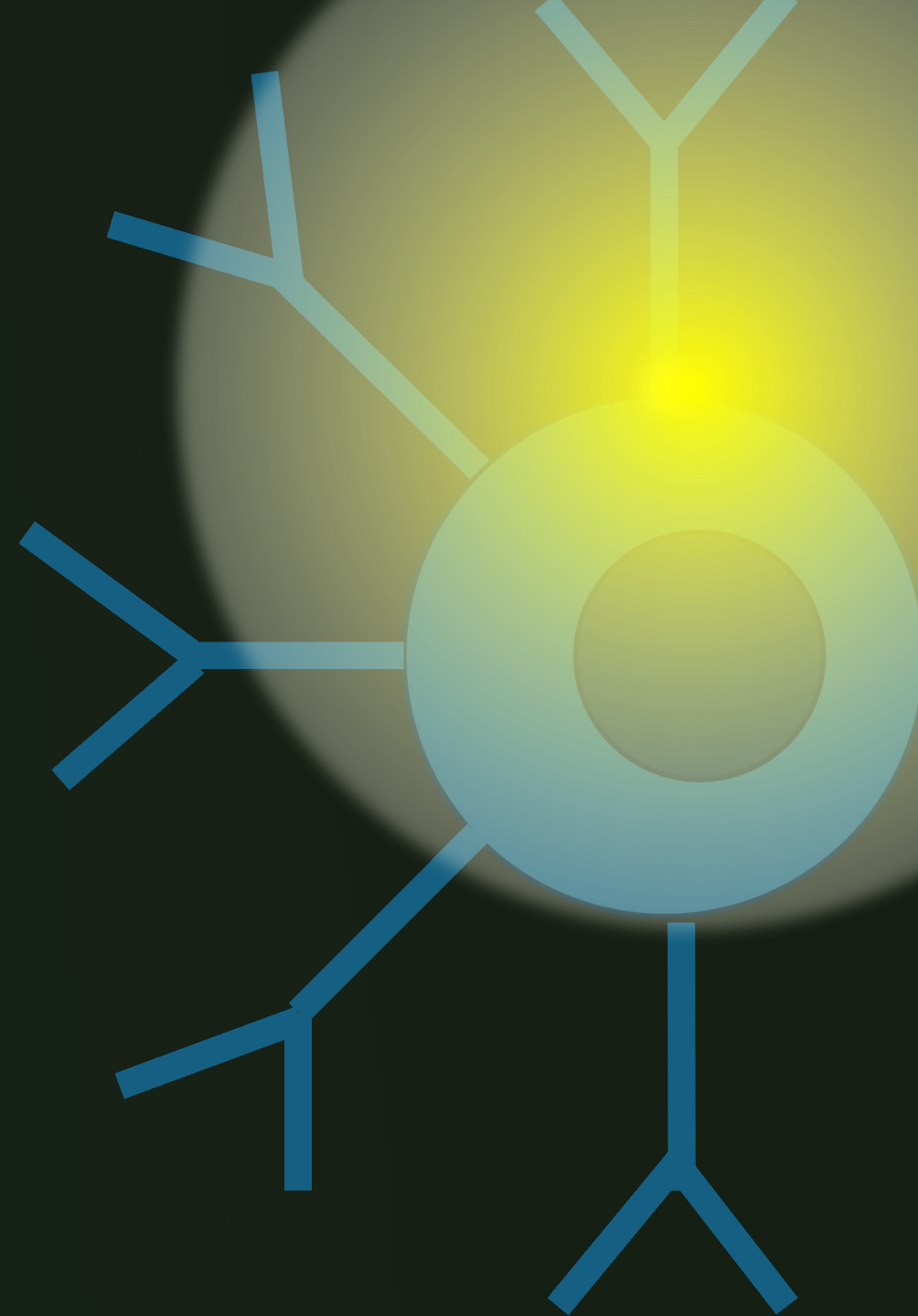
Graded Potentials

@dendrites + soma

Signals die out across distance. Need strong depolarization/hyperpolarization to be carried far

More open channels/more extracellular ligand = stronger signal

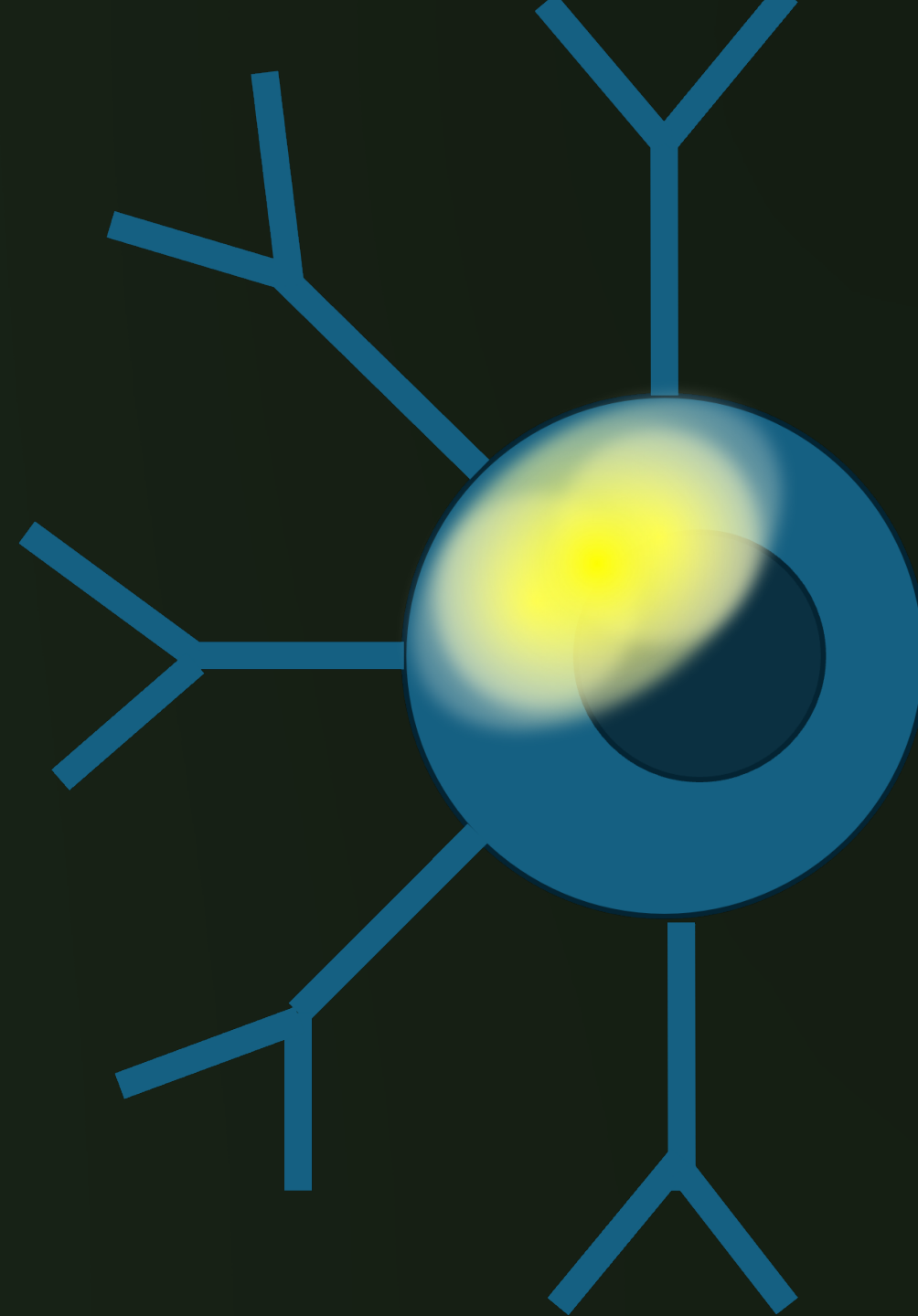
Signal needs to make it to AXON HILLOCK to trigger neuron firing



Spatial Summation

If two (other) neurons synapse on a neuron at the same time (in different places) the potentials will stack

IPSPs and EPSPs will cancel out;
Two EPSPs together will have a higher chance to trigger neuronal firing, and each IPSP lowers the chance of firing.

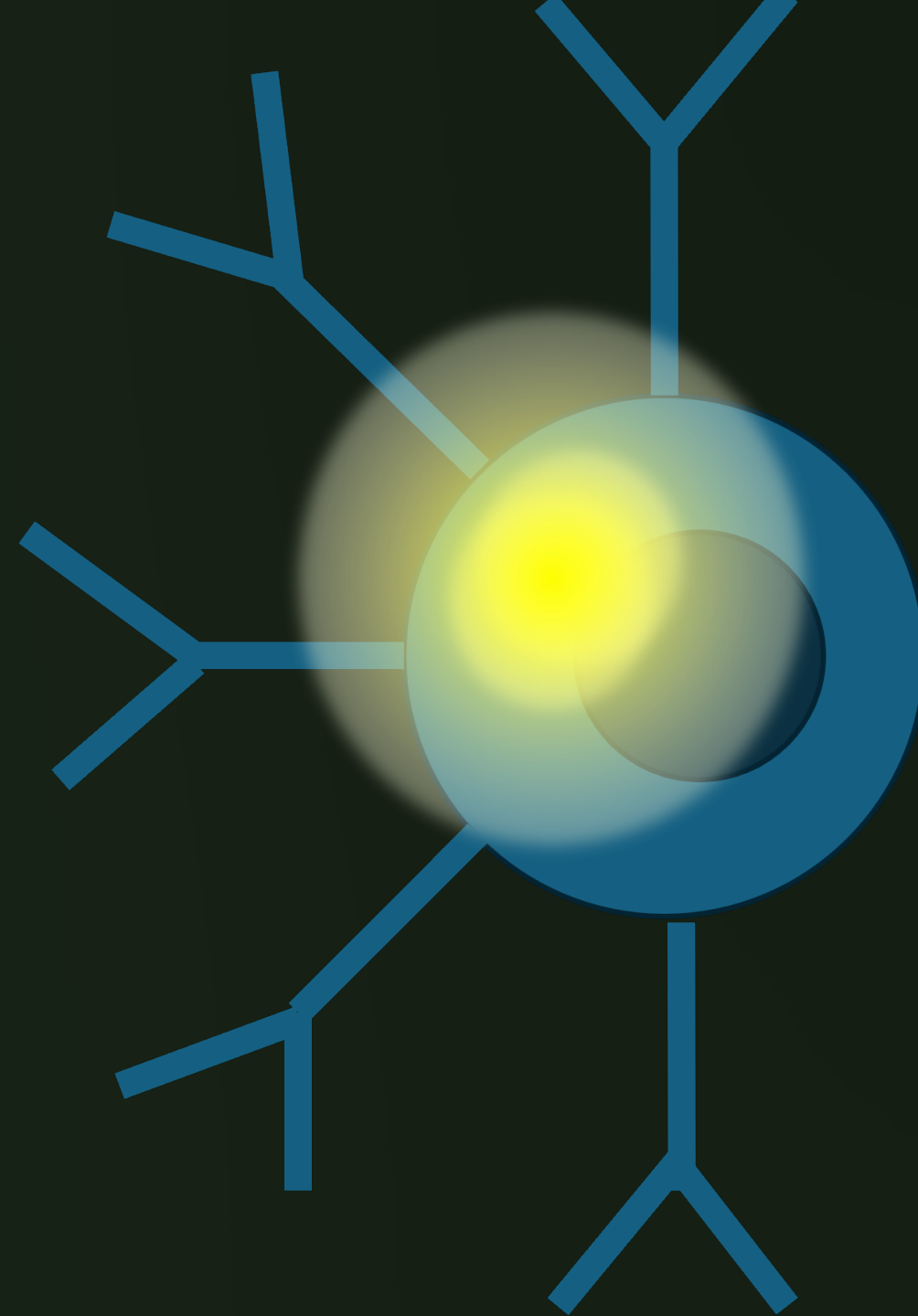


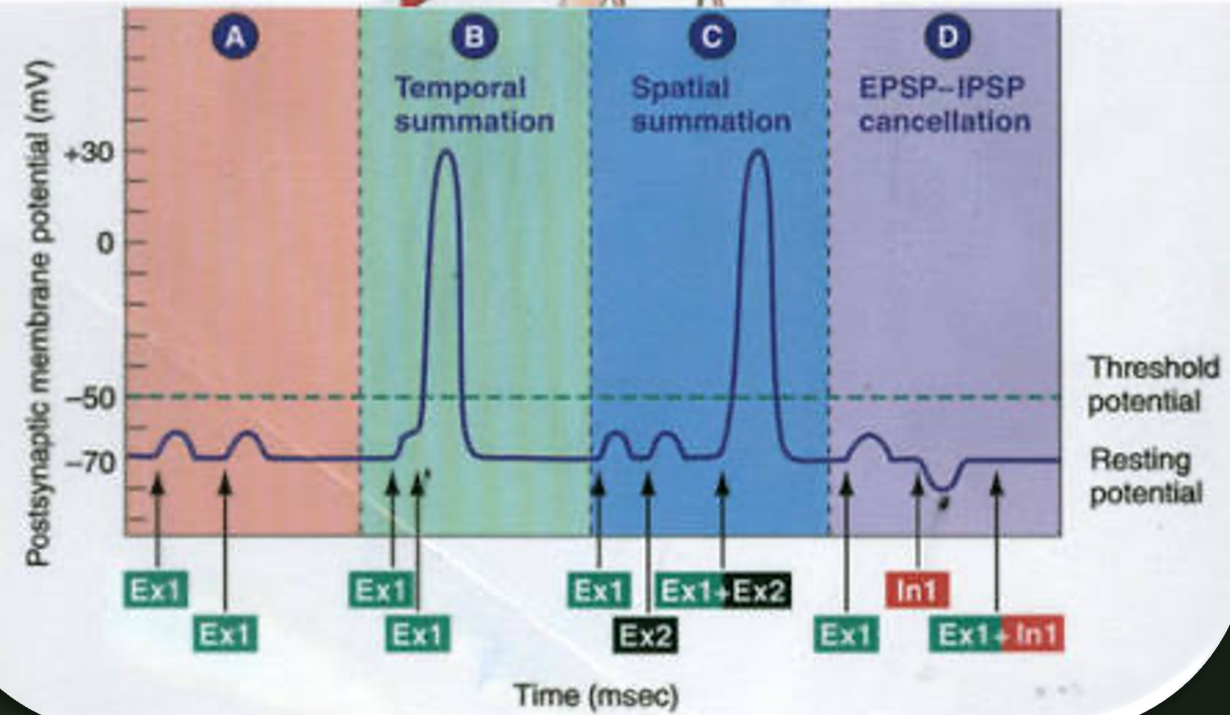
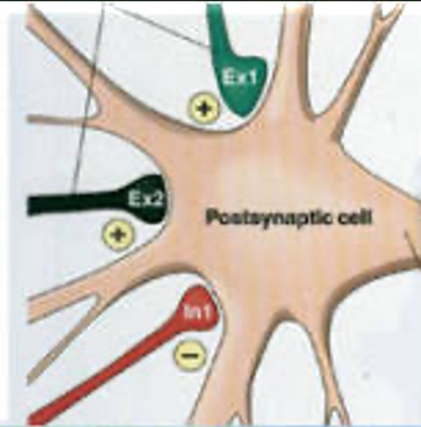
Temporal Summation

If one (other) neuron fires on THIS neuron twice, in short succession, the first potential will not fully return to normal before the second potential; the potentials will stack.

AGAIN IPSPs and EPSPs will cancel out;
Two EPSPs together will have a higher chance to trigger neuronal firing, and each IPSP lowers the chance of firing.

Rapid firing greatly increases the effect (for both EPSPs and IPSPs)





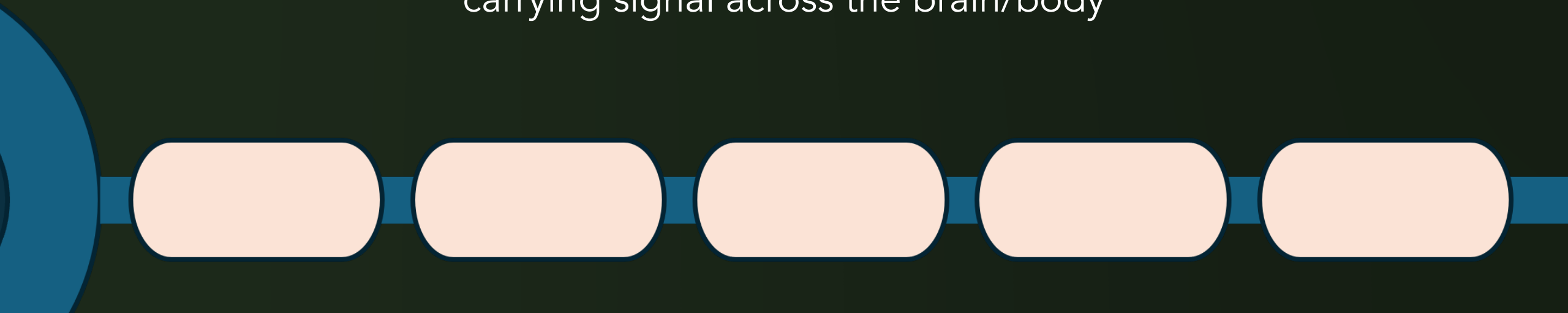
Axon Hillock and Axon

Axon Hillock is the threshold; strong enough EPSPs that reach the hillock will trigger the firing of the neuron

Firing of Neuron is called ACTION POTENTIAL

Action potential starts at hillock, is carried across the axon towards the axon terminal; the axon serves as one big transmission wire

Action potential starts at hillock, carried across the axon towards the axon terminal; the axon serves as one big transmission wire, carrying signal across the brain/body



Action Potential

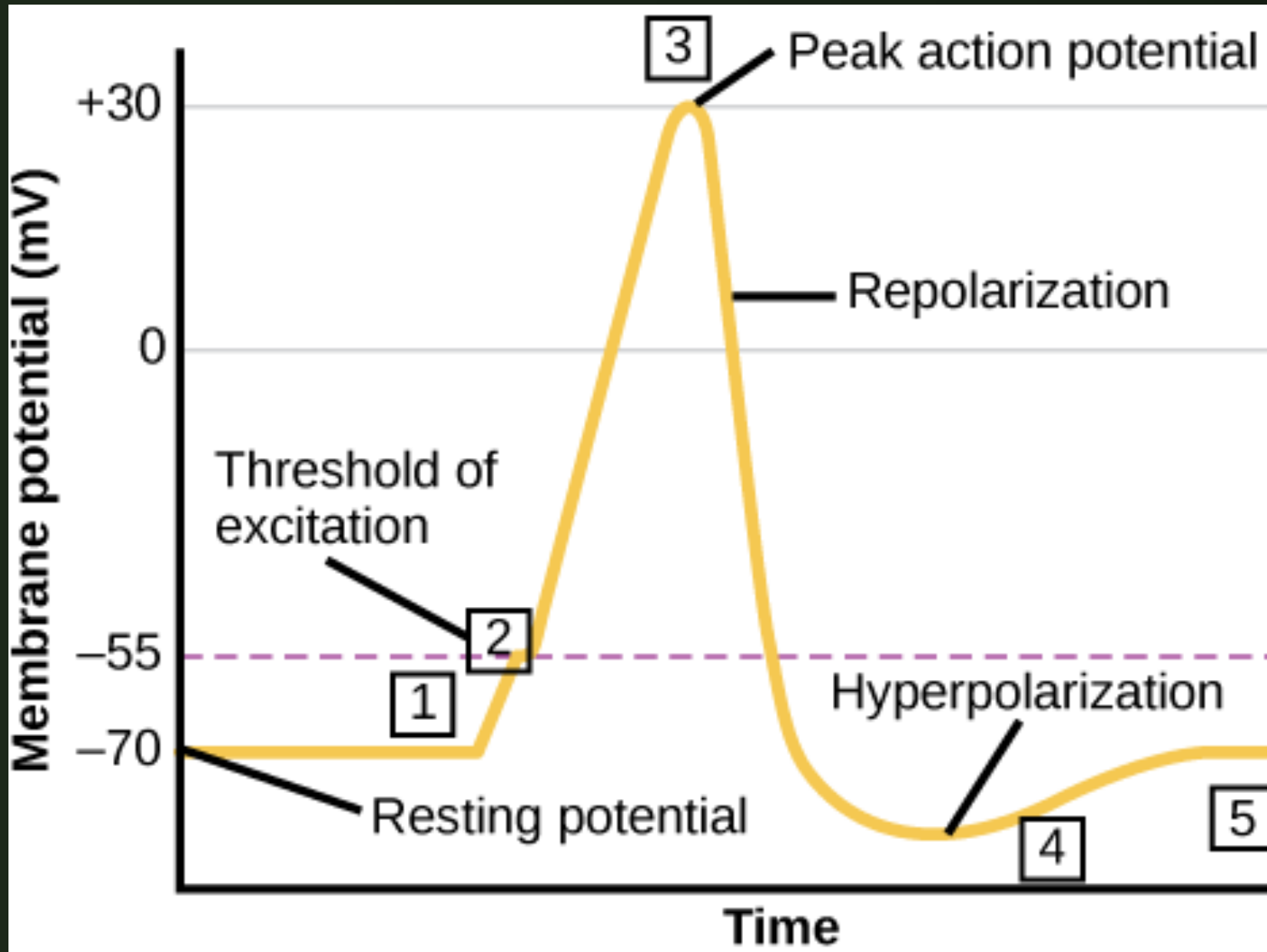
Starting at Axon hillock, the membrane contains high amounts of voltage-gated ion channels. These respond to membrane potential changes and open, ...changing the membrane potential even more.

Unlike in the dendrite or soma, the axon and axon hillock can change the membrane potential.

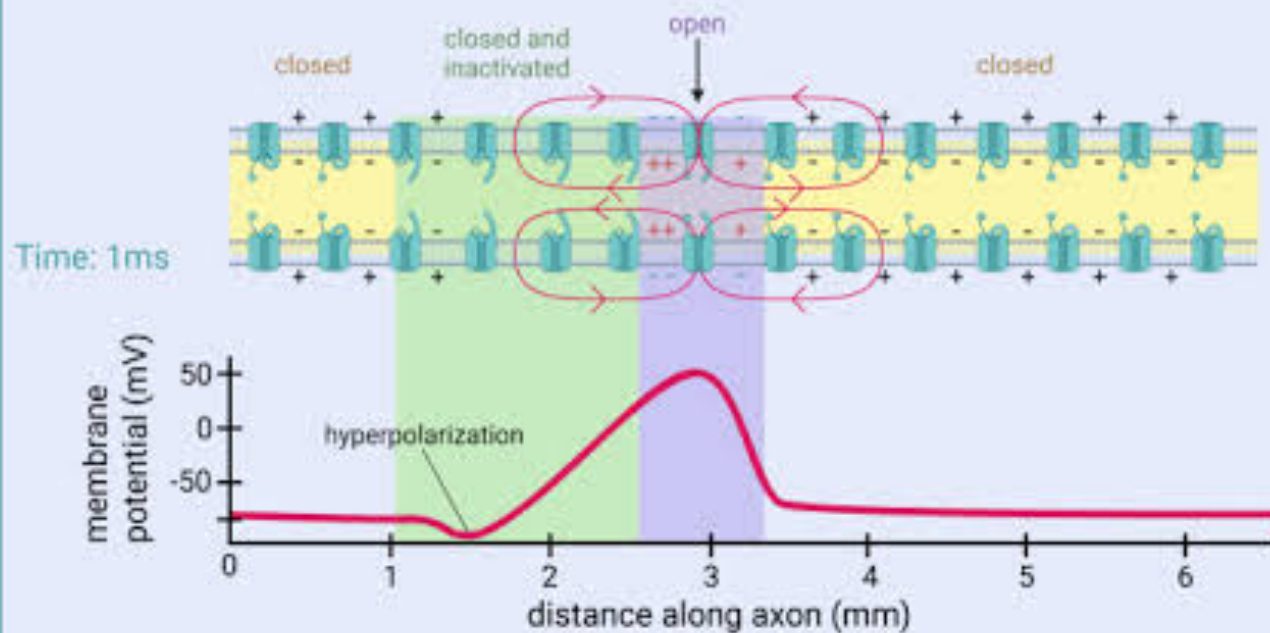
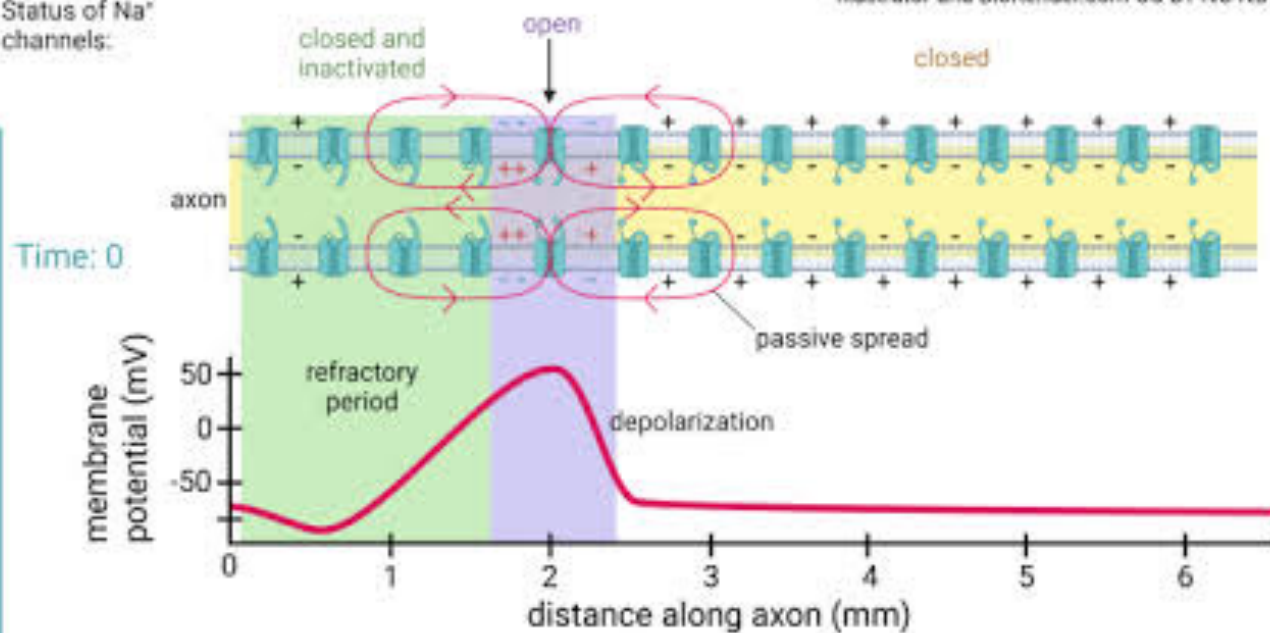
Principle: if membrane potential changes enough to activate voltage-gated ion channels, an action potential will fire and propagates all of the way . If potential doesn't reach this "threshold" potential, firing does not occur

Axon contains two voltage-gated ion channels to propagate the action potential: Voltage-gated Na^+ Channels and Voltage-gated K^+ Channels

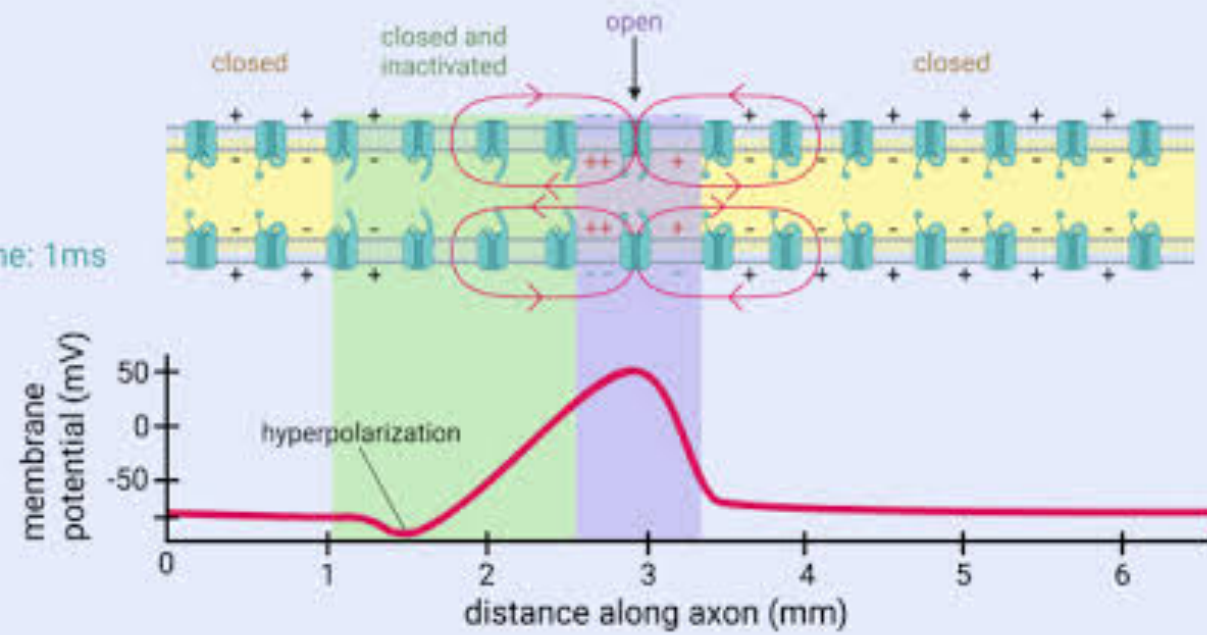
Action Potential (Nihal



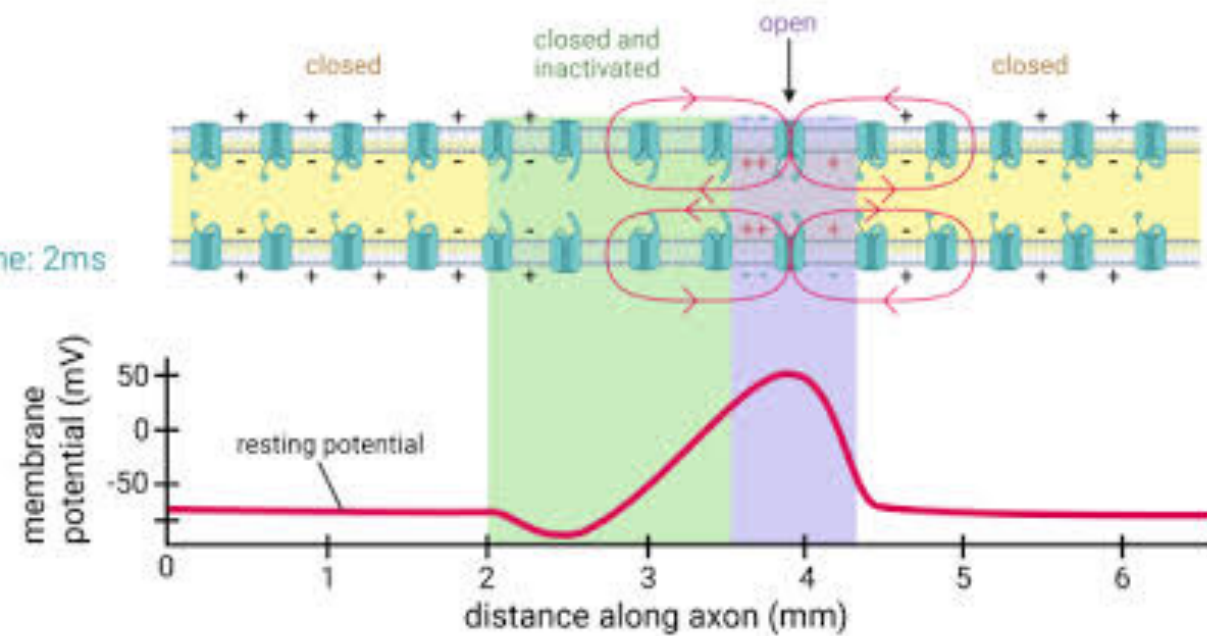
Status of Na⁺
channels:



Time: 1ms



Time: 2ms



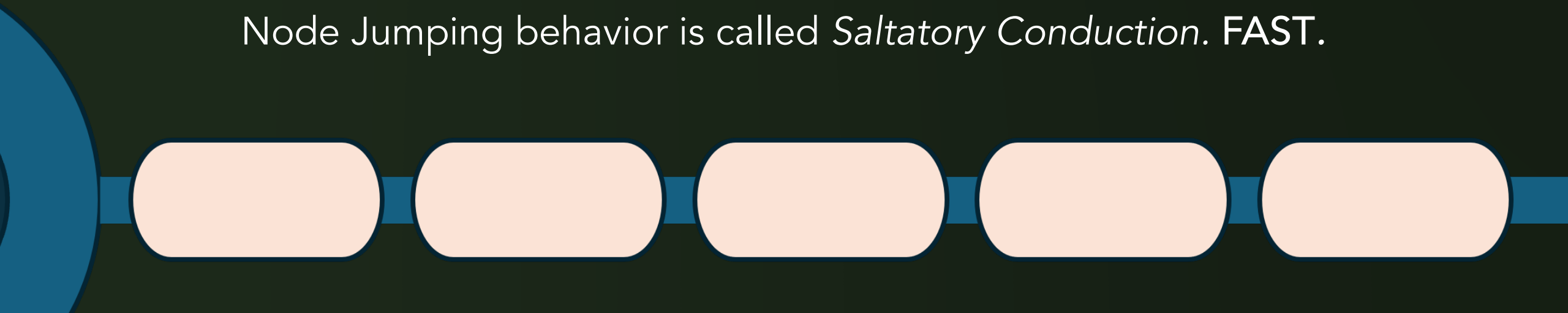
Myelin and Nodes of Ranvier

It takes a lot of charge to trigger an action potential in the next (adjacent) region of the membrane, and membrane charging/gate conformational change delays slow this process

Myelination covers membrane, so ion channels are found densely

packed in Nodes of Ranvier
At each node, depolarization causes an action potential, which pours cations through the membrane. Between nodes (under myelin), electric force pushes intracellular positive ions toward the next node, causing next depolarization

Node Jumping behavior is called *Saltatory Conduction*. **FAST.**

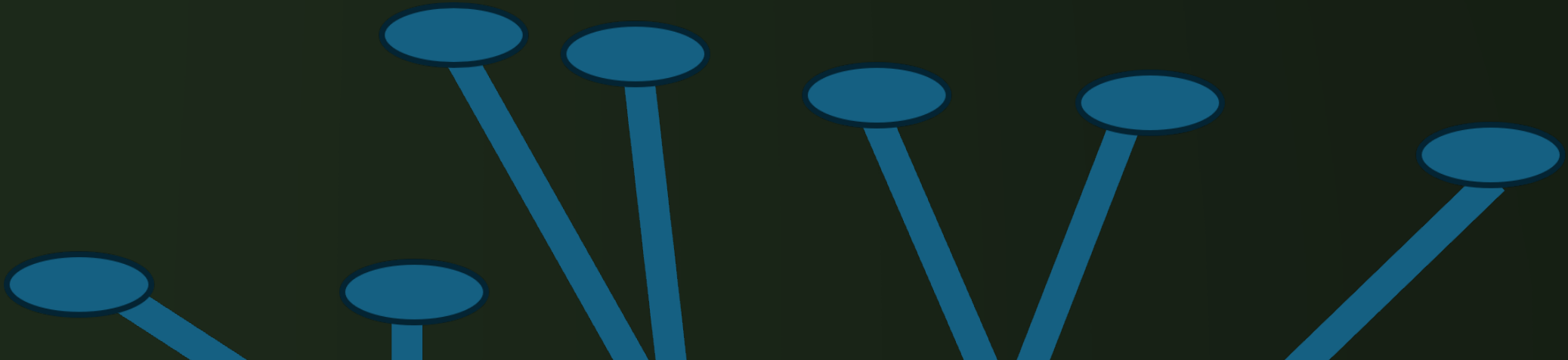


Axon Terminal

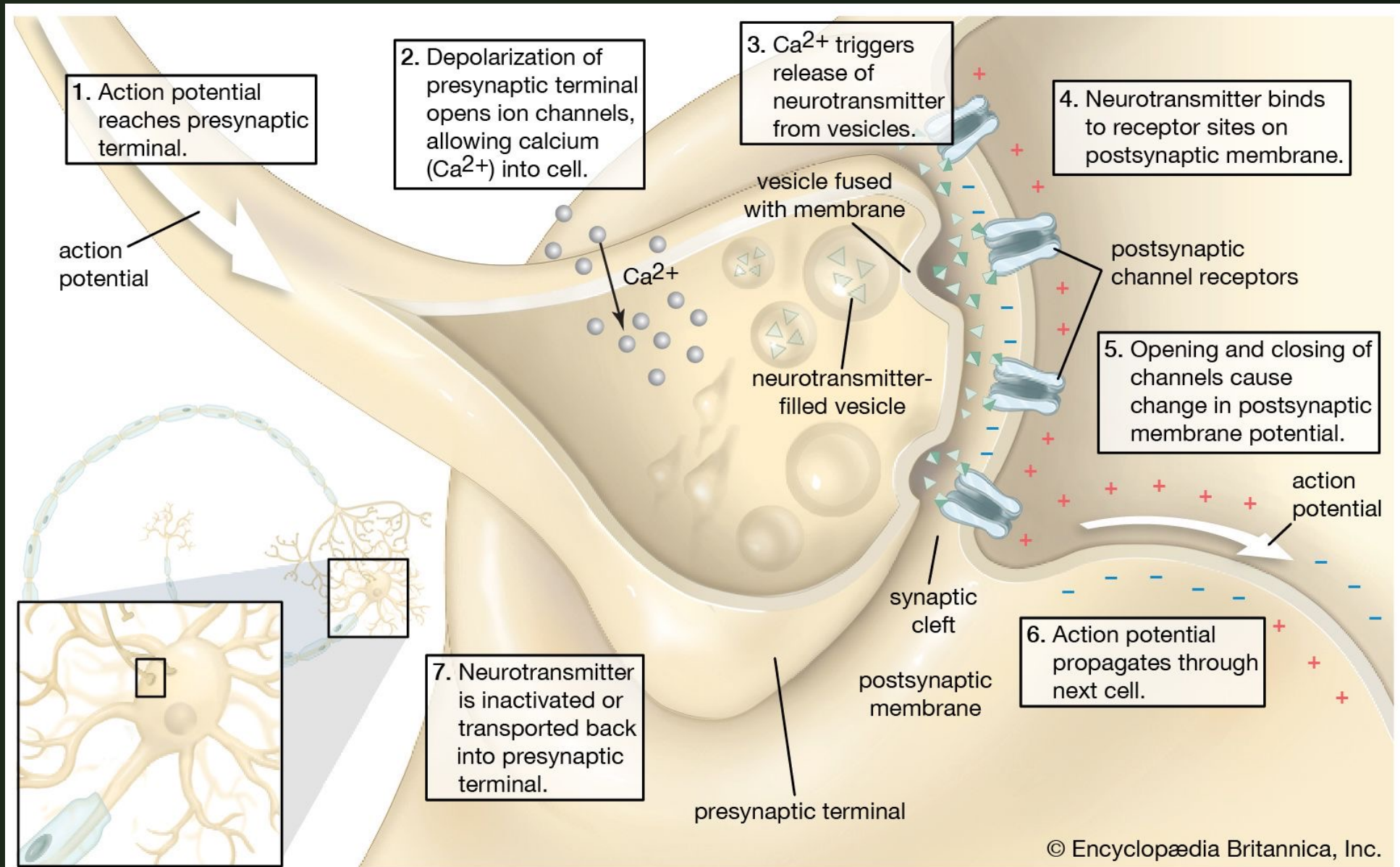
Synaptic boutons contain vesicles – vesicles contain neurotransmitters

Each bouton rests by a synapse, open space. After the synapse is a dendrite of another neuron

Release of NT onto dendrite activates ligand-gated channel:
induces EPSP or IPSP



Axon Terminal Release



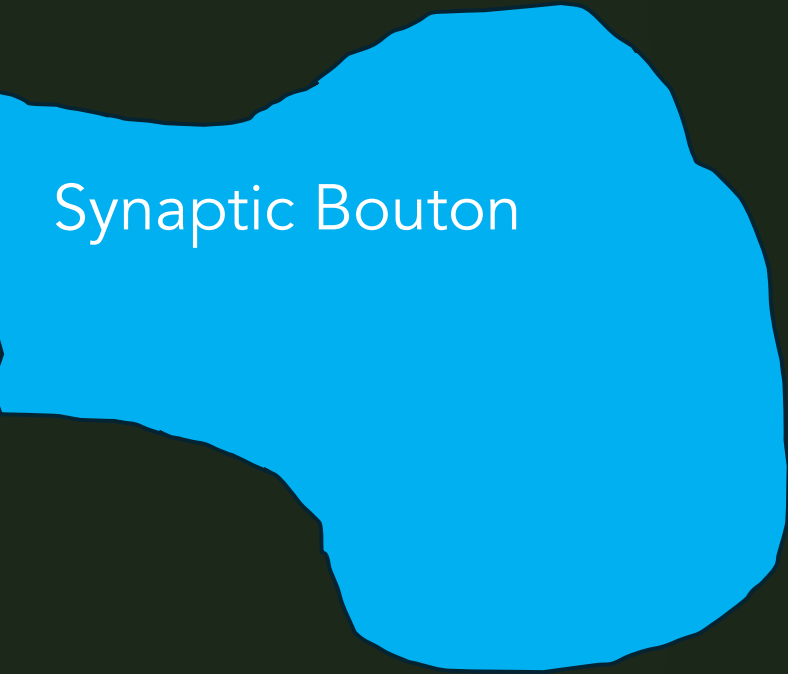
Neurotransmitter Clearance 1



Neurotransmitter Clearance 2

Synaptic Bouton

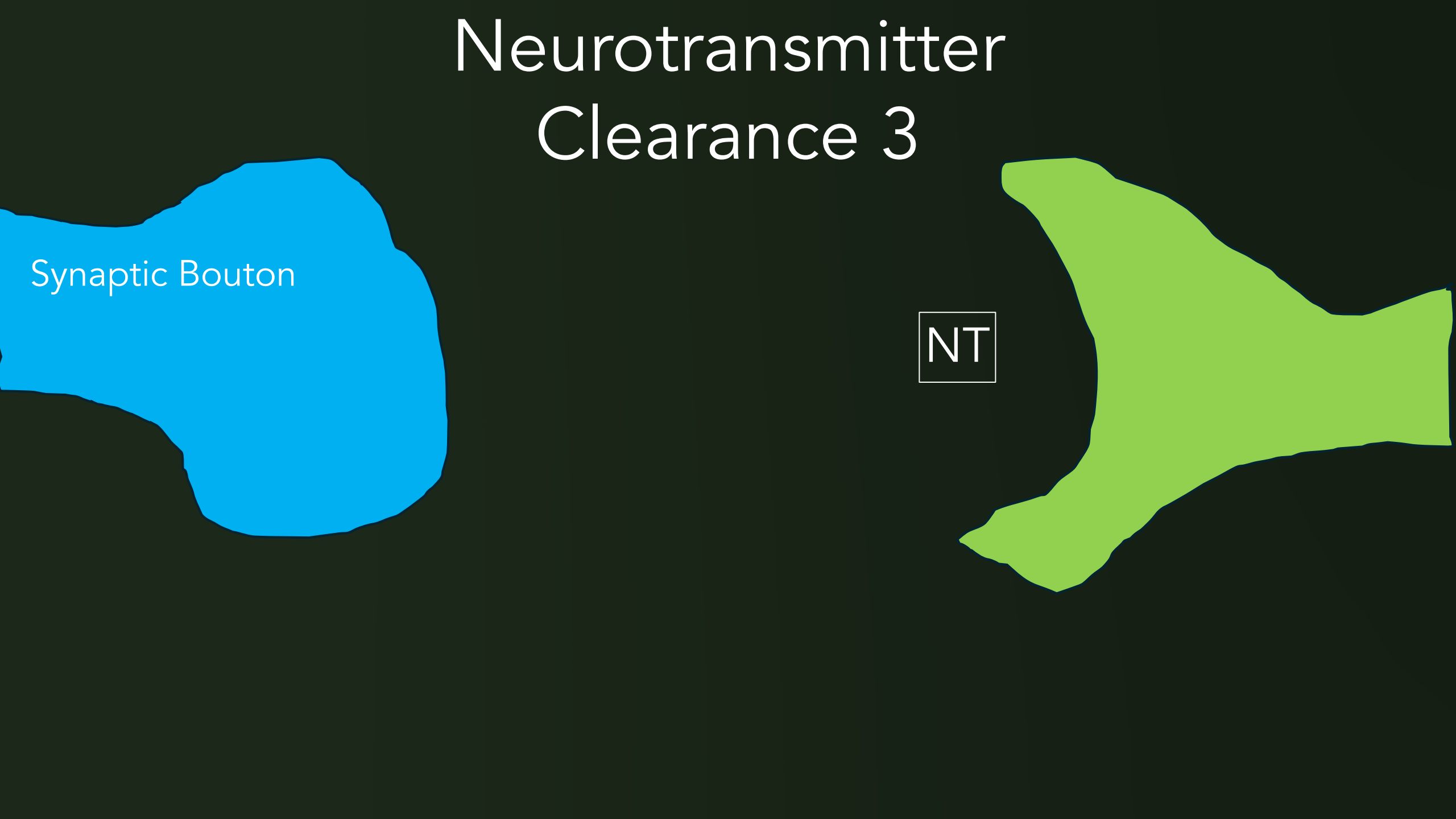
NT



Neurotransmitter Clearance 3

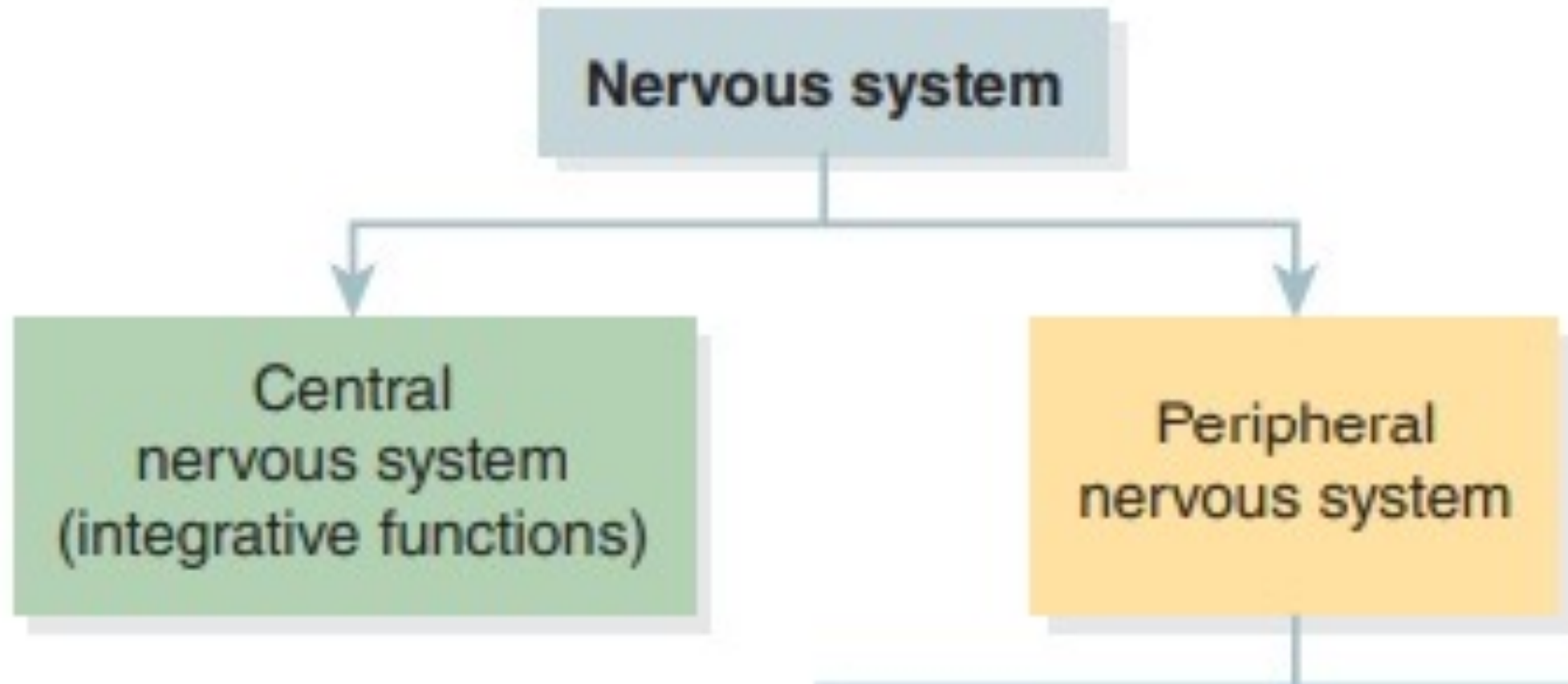
Synaptic Bouton

NT

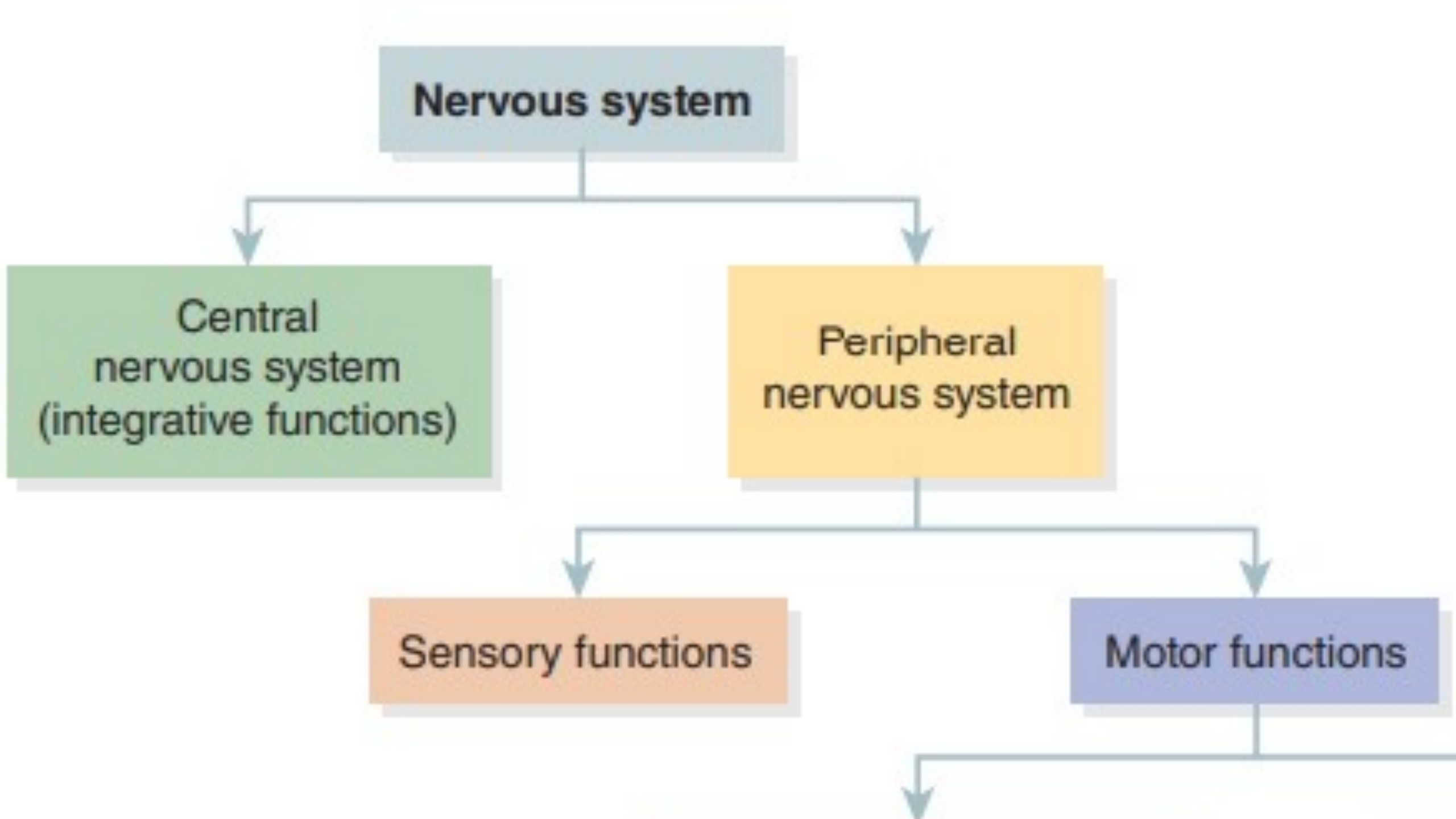
A diagram illustrating a third mechanism for neurotransmitter clearance. On the left is a blue, irregularly shaped 'Synaptic Bouton'. On the right is a green, irregularly shaped structure. Between them is a small white square containing the letters 'NT'. The green structure has a narrow channel or pore that appears to be positioned to take up the 'NT' from the synaptic space.

4.3

Organization of the Nervous System



Nervous system



```
graph TD; A[Nervous system] --> B[Central nervous system<br/>(integrative functions)]; A --> C[Peripheral nervous system]; C --> D[Sensory functions]; C --> E[Motor functions]; E --> F[ ];
```

The diagram is a hierarchical flowchart showing the components of the nervous system. At the top is a light blue box labeled 'Nervous system'. A horizontal line below it branches into two arrows pointing down to 'Central nervous system (integrative functions)' (green box) and 'Peripheral nervous system' (yellow box). From the 'Peripheral nervous system' box, another horizontal line branches into two arrows pointing down to 'Sensory functions' (orange box) and 'Motor functions' (purple box). A final arrow points down from the 'Motor functions' box to an empty space at the bottom of the diagram.

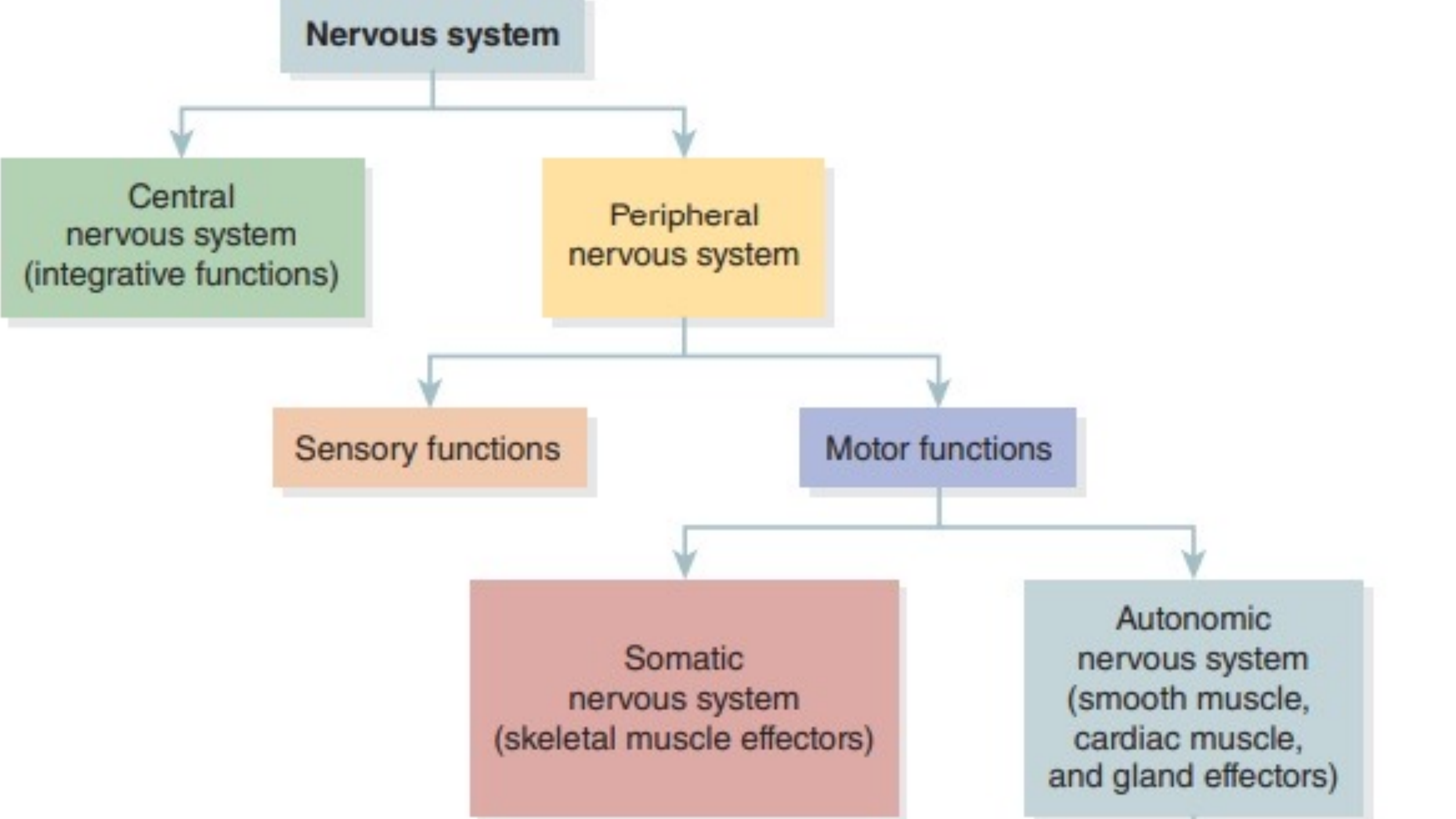
Central
nervous system
(integrative functions)

Peripheral
nervous system

Sensory functions

Motor functions

Nervous system



```
graph TD; A[Nervous system] --> B[Central nervous system<br/>(integrative functions)]; A --> C[Peripheral nervous system]; C --> D[Sensory functions]; C --> E[Motor functions]; E --> F[Somatic nervous system<br/>(skeletal muscle effectors)]; E --> G[Autonomic nervous system<br/>(smooth muscle, cardiac muscle,<br/>and gland effectors)];
```

The diagram is a hierarchical flowchart showing the components of the nervous system. At the top is the 'Nervous system' box. It branches into 'Central nervous system (integrative functions)' and 'Peripheral nervous system'. The 'Peripheral nervous system' further branches into 'Sensory functions' and 'Motor functions'. Finally, 'Motor functions' branches into 'Somatic nervous system (skeletal muscle effectors)' and 'Autonomic nervous system (smooth muscle, cardiac muscle, and gland effectors)'.

Central
nervous system
(integrative functions)

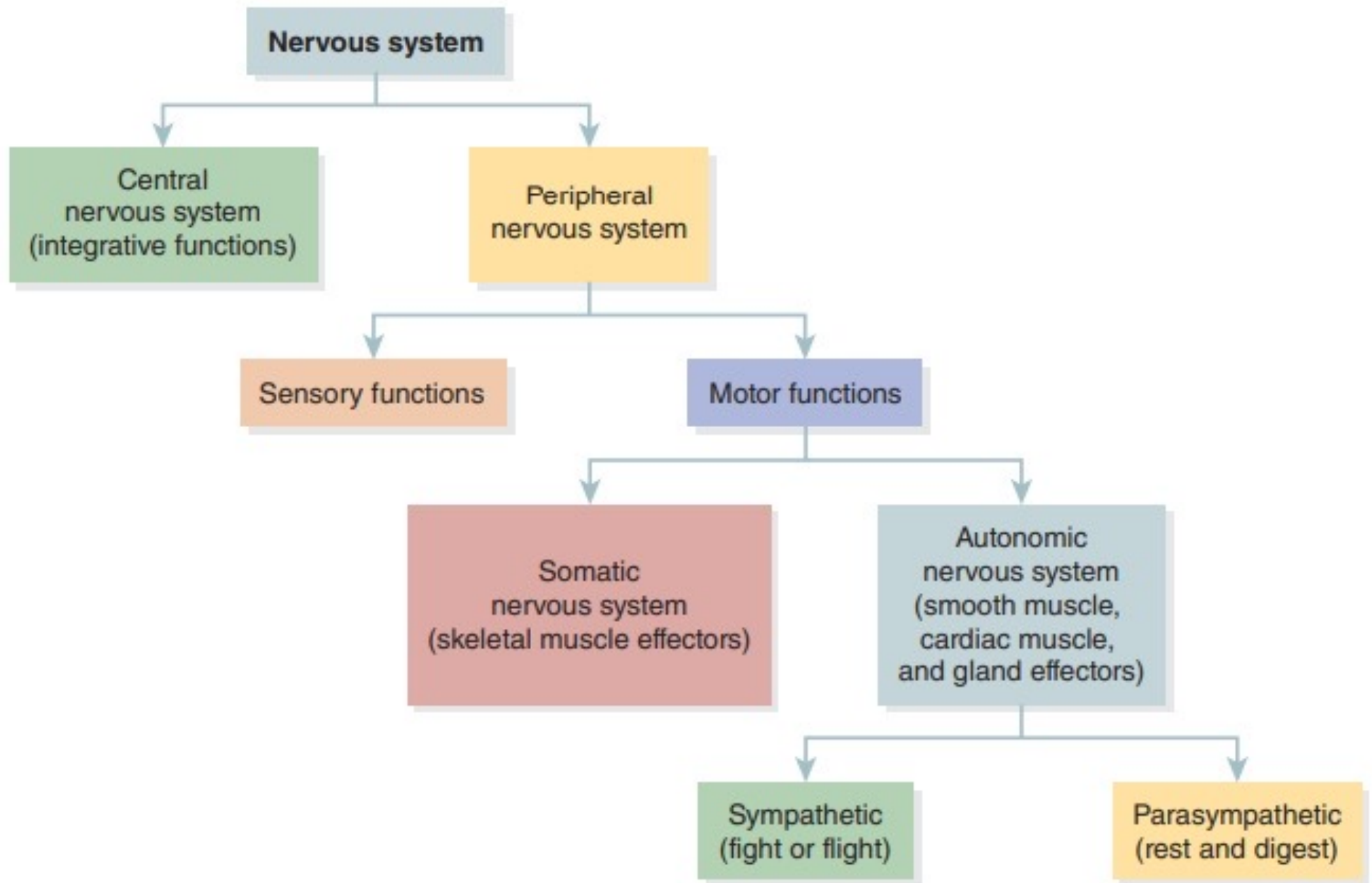
Peripheral
nervous system

Sensory functions

Motor functions

Somatic
nervous system
(skeletal muscle effectors)

Autonomic
nervous system
(smooth muscle,
cardiac muscle,
and gland effectors)



PARASYMPATHETIC NERVES

Vs

SYMPATHETIC NERVES

Constrict
pupils



Eye

Stimulate
Saliva



Pancreas

Slow
heartbeat



Heart

Constrict
airways



Lung



Eye

Dilate
pupils



Pancreas

Inhibit
Salivation



Heart

Increase
heartbeat



Lung

Relax
airways

Stimulate
activity of
Stomach



Stimulate
gallbladder



Stimulate
activity of
Intestines



Contract
bladder



Inhibit
activity of
Stomach



Inhibit
gallbladder



Inhibit
activity of
Intestines



Secrete
epinephrine &
norepinephrine



Relax
Bladder



